

Finite automata on infinite objects

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Finite automata on infinite objects *

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Abstract

Infinite behaviors of several kinds of finite automata on infinite objects are studied. Namely, alternating finite automata on ω -words, finite tree automata on infinite trees which move from the root infinitely, and finite biautomata on two-sided infinite words are investigated through six types of acceptance conditions. Each condition is an appropriate extension of the one used for acceptance of finite automata on ω -words.

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Introduction

This thesis studies infinite behaviors of several kinds of finite automata on infinite objects. Namely, alternating finite automata on ω -words, finite tree automata on infinite trees which move from the root infinitely, and finite biautomata on two-sided infinite words are investigated through six types of acceptance conditions. Each condition is an appropriate extension of the one used for acceptance of finite automata on ω -words.

An ω -word is an infinite sequence of symbols of finite kinds and a set consisting of ω -words is called an ω -language. Originated in Büchi [2], ω -languages have received considerable attentions through various infinite computation models [4–10, 12–15, 18, 20–30, 32–34]. In particular, nondeterministic and deterministic finite automata on ω -words have been extensively studied through a variety of acceptances and the corresponding ω -language classes have been precisely determined [2, 4, 9, 12, 15, 18, 20–21, 24, 29, 32–34]. By these studies six types of acceptance conditions are established as standard for acceptances of ω -words by finite automata. Moreover, they are shown to be sufficient to classify ω -language classes defined by finite automata [18, 32–34]. It is known, in contrast with the case of finite length words, that for two of these six types of acceptance conditions nondeterministic finite automata define larger classes of ω -languages than deterministic ones. The reader is suggested to look up the table in [34, p. 129] for details.

By the discussion above it is natural to use appropriate extensions of these six types of acceptance notions for studying infinite behaviors of several kinds of finite automata on infinite objects. The main aim of

this thesis is to investigate how determinism, nondeterminism or alternation affects the recognition power of infinite objects, i.e. ω -words, infinite trees and two-sided infinite words by alternating finite automata, finite tree automata and finite biautomata, respectively, under these six types of acceptance conditions.

Chapter 1 deals with finite automata on ω -words. This chapter reviews known results concerning nondeterministic and deterministic finite automata on ω -words which are relevant to successive chapters. As an application of infinite behaviors of finite automata we analyze the languages described by flow expressions without synchronization mechanisms. The flow expressions, used to describe flows of sequential and concurrent processes in software descriptions [31], are defined as regular expressions augmented with operators $\parallel$ (shuffle), $\dagger$ (shuffle closure) and ω (infinite repetition). It is shown that the class of languages defined by the flow expressions with $\cdot$, $*$, $\cup$, $\parallel$ and ω coincides with that of infinitary regular languages [24]. We also show that the shuffle closure operator increases the denotational power of flow expressions with the infinite repetition operator.

Alternating finite automata on ω -words are treated in Chapter 2. The main part of this chapter is due to the paper [22]. As an extended notion of nondeterminism, alternation has been studied in relation to the complexity theory [3, 19]. We introduce alternation to finite automata on ω -words. The classes of ω -languages defined by alternating finite automata are investigated under six types of acceptance conditions. Among these six classes we can characterize four classes. For characterization of remaining two classes we only get partial results. Therefore full characterization of these two classes is an open problem.

Chapter 3 is concerned with finite tree automata on infinite trees introduced by Rabin [27–29]. The content of this chapter is based on [14]. Nondeterministic and deterministic finite tree automata on infinite trees which move from the root infinitely are investigated through six types of acceptance conditions. We relate the six classes of infinite tree languages recognized by nondeterministic (resp. deterministic) finite tree automata with the corresponding classes of ω -languages defined by nondeterministic (resp. deterministic) finite automata. The relationship among these six classes is also established.

Finite biautomata on two-sided infinite words are treated in Chapter 4. We define a two-sided infinite word as the equivalence class under the shift of a biinfinite word (i.e. a mapping from the set of integers $\mathbb{Z}$ into a finite alphabet Σ). A finite biautomaton is a pair of finite automata, one of which runs leftward infinitely while the other runs rightward infinitely starting at some point of a two-sided infinite word. The notion of finite biautomata was introduced by Nivat and Perrin [25]. They defined nondeterministic and deterministic finite biautomata through Büchi-type acceptance condition [2]. We consider nondeterministic and deterministic finite biautomata on two-sided infinite words under six types of acceptance conditions including Büchi's one. The classes of two-sided infinite languages defined by finite biautomata are examined in this chapter.

Chapter 1. Finite automata on ω -words

1.1. Introduction

The first half of this chapter reviews known results concerning non-deterministic and deterministic finite automata on ω -words which are relevant to Chapters 2-4. In Section 1.2 we define six types of acceptance conditions for finite automata on ω -words. Each condition uses a family of final sets for acceptance as was done in [21]. So called Büchi-type acceptance [2] and Muller-type acceptance [21] are included in these six types of acceptances.

Characterization of the classes of ω -languages defined by nondeterministic and deterministic finite automata under six types of acceptance conditions are summarized in Section 1.3. Closure properties of these classes are also mentioned. The class of ω -languages defined by deterministic (resp. nondeterministic) finite automata under Muller-type (resp. Büchi-type) acceptance is called the class of ω -regular languages [21]. This class is the most important one.

As an application of infinite behaviors of finite automata we analyze the languages described by flow expressions without synchronization mechanisms in Section 1.4. Flow expressions are introduced by Shaw [31] to describe flows of sequential and concurrent processes. The flow expressions are defined as regular expressions augmented with operators shuffle, shuffle closure and infinite repetition. The infinite repetition op-

erator is used to describe inherently nonterminating flows. The shuffle operator describes the concurrent executions of processes. On the other hand, the shuffle closure is used to describe a variable number of interleaves of some flow.

Formal properties of the languages defined by the flow expressions without the infinite repetition operator have been studied in the literatures [1, 11, 16]. If the flow expressions are allowed the synchronization mechanisms such as *locks* and *waits/signals*, then any recursively enumerable set can be described [1, 16]. It is also known that if no synchronization mechanism is allowed, then the languages described by the flow expressions without the infinite repetition operator are context-sensitive [11].

On the other hand, if the infinite repetition operator is allowed, then the languages described by such flow expressions may contain both finite length words and infinite length words. A language which may contain both finite length words and infinite length words is called an infinitary language [24]. We say that an infinitary language is regular if the part consisting of infinite length words is ω -regular and the remaining part is regular. The class of infinitary regular languages has played an important role in [24]. We prove that the flow expressions without shuffle closure give the class of infinitary regular languages. It is known that the shuffle closure operator increases the denotational power of flow expressions if the infinite repetition operator is not allowed. We show that the same fact also holds for ω -languages defined by the flow expressions with the infinite repetition operator.

1.2. Preliminaries

Let Σ be a finite alphabet. We denote by Σ^ω (resp. Σ^*) the set of all infinite (resp. finite) sequences over Σ . As elements in Σ^* are called *words* and a subset L of Σ^* is called a *language*, elements in Σ^ω are called ω -words and a subset L of Σ^ω is called an ω -language. We denote the empty word by ϵ .

For a set S , $|S|$ denotes the cardinality of S and $P(S)$ is the set of all subsets of S .

DEFINITION 1.2.1. A *nondeterministic finite automaton* (abbreviated *nfa*) is a quintuple $M = \langle S, \Sigma, \delta, s_0, \delta^* \rangle$, where:

- (1) S is a finite set of states.

- (2) Σ is a finite alphabet.
- (3) Δ is a function $\Delta: S \times \Sigma \rightarrow P(S) - \{\emptyset\}$ called the *transition function*.
- (4) s_0 is a state in S called the *initial state*.
- (5) $\mathcal{F}$ is a family of subsets of S . F in $\mathcal{F}$ is called a *final set* and states in F are called *final states*. If $\mathcal{F}$ consists of a single final set, then we say that M is *simple*.

A *deterministic finite automaton* (abbreviated *dfa*) is an *nfa* with the property that $|\Delta(s, \sigma)|=1$ for each s in S and each σ in Σ .

DEFINITION 1.2.2. We can extend the domain of the transition function Δ to $S \times \Sigma^*$ as follows:

For each s in S and x in Σ^* ,

$$\Delta(s, x) = \begin{cases} \{s\} & \text{if } x = \epsilon \\ \bigcup_{q \in \Delta(s, y)} \Delta(q, \sigma) & \text{if } x = y\sigma \text{ for some } y \in \Sigma^* \text{ and } \sigma \in \Sigma. \end{cases}$$

For an *nfa* $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ and x in Σ^* , we say that M *accepts* x if there exists F in $\mathcal{F}$ such that $\Delta(s_0, x) \cap F \neq \emptyset$. We denote by $L(M)$ the set of words accepted by M . A language L is said to be *regular* if there exists an *nfa* M such that $L = L(M)$.

DEFINITION 1.2.3. Let $x = x_0x_1x_2\ldots$ be in Σ^ω , where x_n is in Σ for $n \geq 0$, then a *run* of M on x is a mapping $r: N = \{0, 1, 2, \dots\} \rightarrow S$ such that $r(0) = s_0$ and for each n in N , $r(n+1) \in \Delta(r(n), x_n)$. The set of all runs of M on x is denoted by $Rn(M, x)$. Note that if M is deterministic, then $|Rn(M, x)|=1$ for each x in Σ^ω .

DEFINITION 1.2.4. For a run r in $Rn(M, x)$, We define

- (1) $I(r) = \{s | s \in S, r(i) = s \text{ for infinitely many } i\}$,
- (2) $O(r) = r(N) = \{s | s \in S, r(i) = s \text{ for some } i \in N\}$.

DEFINITION 1.2.5. For a nondeterministic finite automaton $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ and x in Σ^ω , we say that M *accepts* x in the sense of C_i ($i = 1, \dots, 6$) if there exists a run r of M on x such that there is a final set F in $\mathcal{F}$ satisfying the condition C_i , where

- (C₁) $I(r) \cap F \neq \emptyset$,
- (C₂) $I(r) \subseteq F$,
- (C₃) $O(r) \cap F \neq \emptyset$,

- (C₄) $O(r) \subseteq F$,
- (C₅) $I(r) = F$,
- (C₆) $O(r) = F$.

We call r an *accepting run* of M on x in the sense of C_i (or C_i -*accepting run*, in short), respectively. For $i=1, \dots, 6$, we denote by $L_i(M)$ the set of ω -words accepted by M in the sense of C_i and we say that M recognizes an ω -language L in the sense of C_i if $L = L_i(M)$.

The acceptance defined by C_1 condition is sometimes called the acceptance in the sense of Büchi and the C_5 -acceptance is called the acceptance in the sense of Muller.

DEFINITION 1.2.6. For $i=1, \dots, 6$, we define

- (1) $\mathcal{N}_i = \{L_i(M) \mid M \text{ is an nfa}\}$,
- (2) $\mathcal{N}_i^s = \{L_i(M) \mid M \text{ is a simple nfa}\}$,
- (3) $\mathcal{D}_i = \{L_i(M) \mid M \text{ is a dfa}\}$,
- (4) $\mathcal{D}_i^s = \{L_i(M) \mid M \text{ is a simple dfa}\}$.

It is known that a single final set is sufficient under acceptance conditions C_i for $i=1, \dots, 4$ [15, 32]. Namely we have:

THEOREM 1.2.1 [15, 32]. For $i=1, \dots, 4$, $\mathcal{D}_i = \mathcal{D}_i^s$ and $\mathcal{N}_i = \mathcal{N}_i^s$.

Generalizations of this theorem will be treated in successive chapters.

1.3. Overview of the classes concerned

The classes $\mathcal{D}_i$ and $\mathcal{N}_i$ ($i=1, \dots, 6$) have been characterized in terms of general topology and the representations of the ω -languages in these classes have been obtained by applying several operations to regular languages. We list them in the following:

(1) R : An ω -language in R is of the form $\bigcup_{i=1}^n U_i \cdot V_i^\omega$ for some regular languages U_i and V_i , where for $U \subseteq \Sigma^*$ we define $U^\omega = \{y_0 y_1 \dots y_n \dots \mid y_j \text{ is a nonempty word in } U \text{ for } j \geq 0\}$. R is called the class of *ω -regular languages* [21].

(2) G^R : An ω -language in G^R is of the form $U \cdot \Sigma^\omega$, where U is a regular language.

(3) F^R : For a set $U \subseteq \Sigma^*$, $A(U)$ denotes the set $\{w \mid w \in \Sigma^* \text{ and } w \text{ is an initial segment of some word in } U\}$, and we define $U^\omega = \{x \mid x \in \Sigma^\omega \text{ and each initial segment of } x \text{ is in } A(U)\}$. An ω -language in F^R is

described as U^a for some regular language U .

(4) G_δ^R : An ω -language L in G_δ^R is written as $L = W^c$ for some regular language W , where for $W \subseteq \Sigma^*$ we define $W^c = \{x \mid x \in \Sigma^\omega, A(x) \cap W \text{ is infinite}\}$, where $A(x) \subseteq \Sigma^*$ denotes the set of the initial segments of x .

(5) F_σ^R : An ω -language in F_σ^R is of the form $\bigcup_{i=1}^\infty U_i \cdot V_i^c$ for some regular languages U_i and V_i .

We summarize in Table 1-1 the known characterization of the classes of ω -languages defined by deterministic and nondeterministic finite automata on ω -words. We refer the reader to [32, 34] for more details.

Table 1-1

	C_1	C_2	C_3	C_4	C_5	C_6
$\mathcal{D}_1$	G_δ^R	F_σ^R	G^R	F^R	R	$G_\delta^R \cap F_\sigma^R$
$\mathcal{N}_1$	R	F_σ^R	G^R	F^R	R	F_σ^R

In terms of general topology these classes are also characterized as follows: With respect to the product topology on Σ^ω based on the discrete topology on Σ , the classes in Table 1-1 characterized as $G^R = G \cap R$, $F^R = F \cap R$, $G_\delta^R = G_\delta \cap R$ and $F_\sigma^R = F_\sigma \cap R$, where G (resp. F) denotes the class of open (resp. closed) sets and G_δ (resp. F_σ) denotes the class of countable intersections (resp. unions) of open (resp. closed) sets. It is also known that the class of ω -regular languages R is properly included in $G_{\delta\sigma} \cap F_{\sigma\delta}$ [20], where $G_{\delta\sigma}$ (resp. $F_{\sigma\delta}$) denotes the class of countable unions (resp. intersections) of sets in G_δ (resp. F_σ). The inclusion relations among these classes are $G^R \subseteq G_\delta^R \subseteq R$, $F^R \subseteq F_\sigma^R \subseteq R$ and $G^R \cup F^R \subseteq G_\delta^R \cap F_\sigma^R$. Known results concerning these classes which are relevant to successive chapters are summarized as:

THEOREM 1.3.1. [15, 20]. *The classes $G^R \cap F^R$, G^R , F^R , $G_\delta^R \cap F_\sigma^R$, G_δ^R , F_σ^R , and R are pairwise different. Each class is closed under union and intersection.*

THEOREM 1.3.2. [2]. *The class of ω -regular languages R is closed under complement.*

1.4. Relations to flow expression languages

We define $\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$. Any subset of Σ^∞ is called an *infinitary language*. For $L \subseteq \Sigma^\infty$, we define $L^{fin} = L \cap \Sigma^*$ and $L^{inf} = L \cap \Sigma^\omega$. The sets L^{fin} and L^{inf}

are called the *finitary part* and the *purely infinitary part* of L , respectively.

DEFINITION 1.4.1. For infinitary languages L , L_1 and L_2 , the infinitary languages $L_1 \cdot L_2$, L^* , L^ω , $L_1 || L_2$ and L^+ are defined as follows:

- (1) $L_1 \cdot L_2 = \{xy \mid x \in L_1^{fin} \text{ and } y \in L_2\} \cup L_1^{inf}$.
- (2) $L^* = \{\varepsilon\} \cup L \cup L \cdot L \cup L \cdot L \cdot L \cup \dots$
- (3) $L^\omega = \{x_0 x_1 \dots x_j \dots \in \sum^\omega \mid x_j \in L^{fin} - \{\varepsilon\}\} \cup (L^{fin})^* \cdot L^{inf}$.
- (4) $L_1 || L_2 = \{x_0 y_0 \dots x_j y_j \dots \mid x_0 \dots x_j \dots \in L_1, y_0 \dots y_j \dots \in L_2, x_j, y_j \in \sum^*\}$,

where we use the convention that two dots terminating a string denotes both finite and infinite length words.

- (5) $L^+ = \{\varepsilon\} \cup L \cup L || L \cup L || L || L \cup \dots$

DEFINITION 1.4.2. *Flow expressions* (FE's) over an alphabet Σ are defined as follows: Each σ in Σ is an FE. The special symbols λ and ϕ are FE's. If E , E_1 and E_2 are FE's, then so are $E_1 \cdot E_2$ (*concatenation*), $E_1 \cup E_2$ (*union*), E^* (*sequential closure*), E^ω (*infinite repetition*), $E_1 || E_2$ (*shuffle*), and E^+ (*shuffle closure*). The *flow language* $L(E)$ associated with a flow expression E is defined in the following manner:

- (1) $L(\sigma) = \{\sigma\}$ for $\sigma \in \Sigma$.
- (2) $L(\lambda) = \{\varepsilon\}$.
- (3) $L(\phi) = \phi$ (empty set).
- (4) $L(E_1 \cdot E_2) = L(E_1) \cdot L(E_2)$.
- (5) $L(E_1 \cup E_2) = L(E_1) \cup L(E_2)$.
- (6) $L(E^*) = L(E)^*$.
- (7) $L(E^\omega) = L(E)^\omega$.
- (8) $L(E_1 || E_2) = L(E_1) || L(E_2)$.
- (9) $L(E^+) = L(E)^+$.

For a given set of operators $X \subseteq \{ \cdot, *, \cup, \omega, ||, + \}$, $FE(X)$ denotes the class of languages defined by the flow expressions using the constructors in X . Thus $FE(\cdot, *, \cup)$ represents the class of regular languages.

We define the class of *infinitary regular languages* over a finite alphabet Σ as:

$R(\Sigma^\infty) = \{L \mid L^{fin} \text{ is a regular language and } L^{inf} \text{ is an } \omega\text{-regular language}\}.$

Now we can show relations between infinitary regular languages and flow expressions without shuffle closure.

THEOREM 1.4.1. *The following classes are identical.*

- (1) $R(\Sigma^\infty)$.
- (2) $FE(\cdot, *, \cup, \omega)$.
- (3) $FE(\cdot, *, \cup, \omega, ||)$.
- (4) $FE(\cdot, *, \cup, \cap, \omega, ||)$.

PROOF. We first show that the equality $R(\Sigma^\infty) = FE(\cdot, *, \cup, \omega)$. For L in $R(\Sigma^\infty)$, its finitary part L^{fin} is regular and its purely infinitary part L^{inf} is ω -regular, and is written as

$$L^{inf} = \bigcup_{i=1}^n U_i \cdot V_i^\omega,$$

where U_i and V_i are regular languages. (Cf. Section 1.3.) Thus we can construct a flow expression representing L by using $\cdot$, $*$, $\cup$ and ω . Hence $R(\Sigma^\infty) \subseteq FE(\cdot, *, \cup, \omega)$.

To prove the converse, it is sufficient to show that $R(\Sigma^\infty)$ is closed under the operators $\cdot$, $*$, $\cup$ and ω . This can be easily shown and we omit the details.

Next we show the containment $FE(\cdot, *, \cup, \omega, ||) \subseteq R(\Sigma^\infty)$. From the above discussion it suffices to show that if L_1 and L_2 are in $R(\Sigma^\infty)$, then so is $L_1 || L_2$. Since $L_1 || L_2 = L_1^{fin} || L_2^{fin} \cup L_1^{fin} || L_2^{inf} \cup L_1^{inf} || L_2^{fin} \cup L_1^{inf} || L_2^{inf}$ and the class of regular languages is closed under shuffle $||$, it suffices to show that $L_1^{fin} || L_2^{inf}$, $L_1^{inf} || L_2^{fin}$ and $L_1^{inf} || L_2^{inf}$ are ω -regular languages.

First we show that $L_1^{inf} || L_2^{inf}$ is an ω -regular language. Let $M_k = \langle S_k, \Sigma, \Delta_k, s_{0k}, \mathcal{F}_k \rangle$ be a *dfa* such that $L_k^{inf} = L_5(M_k)$ for $k=1, 2$. We construct an *nfa* M such that $L_5(M) = L_5(M_1) || L_5(M_2)$ in the following way: We set

- (1) $S = S_1 \times S_2$ and $s_0 = (s_{01}, s_{02})$,
- (2) $\Delta((s_1, s_2), \sigma) = \{(s_1, s_2) | s_1 = \Delta_1(s_1, \sigma)\} \cup \{(s_1, s_2) | s_2 = \Delta_2(s_2, \sigma)\}$,
- (3) $\mathcal{F} = \{F | F \subseteq S_1 \times S_2, p_1(F) \in \mathcal{F}_1, p_2(F) \in \mathcal{F}_2\}$, where p_1 and p_2 are the projection functions $(x, y) \mapsto x$ and $(x, y) \mapsto y$, respectively.

It is easily shown that $L_5(M) = L_5(M_1) || L_5(M_2)$.

Next we show that $L_1^{fin} || L_2^{inf}$ is an ω -regular language. Let $M_1 = \langle S_1, \Sigma, \Delta_1, s_{01}, \{F_1\} \rangle$ be a *dfa* such that $L_1^{fin} = L(M_1)$ and $M_2 = \langle S_2, \Sigma, \Delta_2, s_{02}, \mathcal{F}_2 \rangle$

be a *dfa* such that $L_2^{inf} = L_s(M_2)$. Construct an *nfa* $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ in the following way:

We set $S = (S_1 \cup \{\#\}) \times S_2$, $s_0 = (s_{01}, s_{02})$ and $\mathcal{F} = \{\#\} \times \mathcal{F}_2$, where $\#$ is a new symbol. The transition function Δ is defined as follows:

- (1) $\Delta((s_1, s_2), \sigma) = \{(s, s_2) \mid s = \Delta_1(s_1, \sigma)\} \cup \{(s_1, s) \mid s = \Delta_2(s_2, \sigma)\}$ if $s_1 \in S_1 - F_1$,
- (2) $\Delta((s_1, s_2), \sigma) = \{(s, s_2) \mid s = \Delta_1(s_1, \sigma)\} \cup \{(\#, s), (s_1, s) \mid s = \Delta_2(s_2, \sigma)\}$ if $s_1 \in F_1$, and
- (3) $\Delta((\#, s_2), \sigma) = \{(\#, s) \mid s = \Delta_2(s_2, \sigma)\}$.

It is easily verified that $L_s(M) = L(M_1) \parallel L_s(M_2)$.

Finally, as is pointed out in [24], $R(\Sigma^\omega)$ is also closed under intersection. Therefore $FE(\cdot, *, \cup, \cap, \omega, \parallel)$ is equal to $R(\Sigma^\omega)$ if flow expressions are extended to include the intersection operator.

REMARK. Closedness of the class of ω -regular languages R under the shuffle is reported by Park [26] and independently by us [13] in which Büchi-type acceptance (C_1 -acceptance) condition is used to define R and the proof is more complicated.

The following proposition asserts that shuffle closure also increases the denotational power for purely infinitary flow languages.

PROPOSITION 1.4.2. *There exists a language $L \subseteq \Sigma^\omega$ such that L is in $FE(\cdot, *, \cup, \omega, \dagger) - FE(\cdot, *, \cup, \omega)$.*

PROOF. Consider $E = ((a \cdot b)^\dagger)^\omega$. If $L(E)$ is in $FE(\cdot, *, \cup, \omega)$, then by Theorem 1.4.1, $L = L(E) \cap (a^* \cdot b^*)^\omega$ is also in $R(\Sigma^\omega)$. But $L = \{a^{n_1} b^{n_2} \dots \mid n_i \geq 1\}$ and it can be easily shown that L is not in $R(\Sigma^\omega)$.

The next theorem is useful to show that certain languages are not in $FE(\cdot, *, \cup, \cap, \parallel, \dagger)$. We use the languages A_m and B_m for $m \geq 0$ defined as follows: $A_0 = a^*$ and $A_m = a^* c A_{m-1}$, where a and c are distinct symbols, and $B_m = A_0 \cup \dots \cup A_m$.

THEOREM 1.4.3. *For any L in $FE(\cdot, *, \cup, \cap, \parallel, \dagger)$, $B_m \cap L$ is regular for $m \geq 0$.*

PROOF. Let E be a flow expression such that $L = L(E)$. If E does not contain the shuffle closure operator $\dagger$, then $L(E)$ is a regular lan-

guage since $FE(\cdot, *, \cup, \cap, ||)$ coincides with the class of regular languages. Therefore $B_m \cap L$ is regular. We show that if the number of occurrences of $\dagger$ in E is k ($k \geq 1$), then we can construct a flow expression E' with $k-1$ occurrences of $\dagger$ from E such that $B_m \cap L(E) = B_m \cap L(E')$. In this construction we make use of the following observations. For a flow expression using only one alphabet symbol, the shuffle closure operator has the same effect as the star operator. Namely, $K^\dagger = K^*$ for $K \subseteq \{a\}^*$. Since the intersection with B_m is taken finally to get the language defined, strings with more than m occurrences of the symbol c in $L(\tilde{E})$, where $\tilde{E}$ is an arbitrary subexpression of E , do not contribute the language finally defined. Only strings in B_m contribute the language finally defined. Let $(\tilde{E})^\dagger$ be a subexpression of E such that $\tilde{E}$ does not involve any $\dagger$ operator. Then $L(\tilde{E})$ is a regular language. For $i = 0, \dots, m$, let E_i be a regular expression which expresses the regular language $L(\tilde{E}) \cap A_i$. Then by the observations made above, the contribution of $(\tilde{E})^\dagger$ to the language finally defined can be replaced by

$$(1) \quad E_0^* \cup E_0^* || \left(\bigcup_{\substack{1 \leq i_1 \leq \dots \leq i_t \\ 1 \leq i_1 + \dots + i_t \leq m}} E_{i_1} || E_{i_2} || \dots || E_{i_t} \right)$$

Therefore the desired expression E' is obtained from E by replacing $(\tilde{E})^\dagger$ by (1).

Since the language $\{a^n c a^n | n \geq 0\}$ is not regular, we have the following corollary:

COROLLARY 1.4.4. *The language $\{a^n c a^n | n \geq 0\}$ is not in $FE(\cdot, *, \cup, \cap, ||, \dagger)$.*

REMARK. Notice that Corollary 1.4.4 cannot be proved by the method shown in [11] (Lemma IV. 7). It cannot be also proved by the pumping lemma in [17] since intersection is allowed. In contrast with Corollary 1.4.4, $\{a^n c b^n | n \geq 0\}$ is in $FE(\cdot, *, \cup, \cap, ||, \dagger)$ because $L((ab \cup c)^\dagger \cap a^* c b^*) = \{a^n c b^n | n \geq 0\}$. Therefore the class $FE(\cdot, *, \cup, \cap, ||, \dagger)$ is not closed under homomorphisms since $\{a^n c a^n | n \geq 0\}$ is the homomorphic image of $\{a^n c b^n | n \geq 0\}$ under the homomorphism h defined by $h(a) = a$, $h(b) = a$ and $h(c) = c$.

1.5. Concluding remarks

In the definition of finite automata we have assumed that $\Delta(s, \sigma) \neq \emptyset$ for each s in S and each σ in Σ . In the literatures [15, 34], the case allowing $\Delta(s, \sigma) = \emptyset$ has also been discussed. We call an *nfa* which allows the case $\Delta(s, \sigma) = \emptyset$ a *partial nfa*, partial *dfa* can be defined similarly. Let $\mathcal{PD}_i, \mathcal{PN}_i$ ($i=1, \dots, 6$) be the classes of ω -language recognized by partial *dfa*'s, partial *nfa*'s in the sense of C_i , respectively. It is not hard to show that the partial and nonpartial classes are equal under all but C_3 type acceptances for deterministic and nondeterministic cases. This can be shown by constructing a nonpartial finite automaton by adding a trapping state into which all undefined transitions come. For C_3 type acceptance, $\mathcal{PD}_3$ and $\mathcal{PN}_3$ have been already determined [15, 34]. Table 1-2 summarizes the classes for partial finite automata on ω -words.

Table 1-2

	C_1	C_2	C_3	C_4	C_5	C_6
$\mathcal{PD}_i$	G_σ^R	F_σ^R	$G^R \Delta F^R$	F^R	R	$G_\sigma^R \cap F_\sigma^R$
$\mathcal{PN}_i$	R	F_σ^R	F_σ^R	F^R	R	F_σ^R

$G^R \Delta F^R = \{L_1 \cap L_2 \mid L_1 \in G^R, L_2 \in F^R\}.$

To characterize the class of infinitary languages $FE(\cdot, *, \cup, \cap, \omega, ||, \dagger)$ is an interesting open problem since $FE(\cdot, *, \cup, \cap, ||, \dagger)$ is a proper subclass of context-sensitive languages by [11] and Corollary 1.4.4.

Chapter 2. Alternating finite automata on ω -words

2.1. Introduction

As an extended notion of nondeterminism, alternation has been introduced and many kinds of alternating automata have been studied in relation to the complexity theory [3, 19]. It has been shown that alternating finite automata on finite length words accept only regular languages whether they have one-way input heads or two-way input heads [3, 19].

Alternation can be also introduced to finite automata on ω -words. When finite automata are viewed as a method of describing the sets of ω -words, alternation has a benefit that it makes the descriptions easy

and simple. The purpose of this chapter is to investigate how alternation affects the recognition power of ω -languages by finite automata in comparison with nondeterministic ones. We define six types of acceptance conditions for alternating finite automata on ω -words in the similar way as done in Chapter 1. The classes of ω -languages defined under four types of acceptance conditions are characterized. Only partial results are obtained for characterization of the classes defined under remaining two types of acceptance conditions.

2.2. Preliminaries

DEFINITION 2.2.1. An *alternating finite automaton* (abbreviated *afa*) is a sextuple $M = \langle S, f, \Sigma, \Delta, s_0, \mathcal{F} \rangle$, where:

- (1) S is a finite set of states and f is a mapping $f: S \rightarrow \{\text{and}, \text{or}\}$. If $f(s) = \text{and}$ (resp. or), s is called a *universal state* (resp. *existential state*).
- (2) Σ is a finite alphabet.
- (3) Δ is a function $\Delta: S \times \Sigma \rightarrow P(S) - \{\emptyset\}$ called the *transition function*.
- (4) s_0 is a state in S called the *initial state*.
- (5) $\mathcal{F}$ is a family of subsets of S . F in $\mathcal{F}$ is called a *final set* and states in F are called *final states*. If $\mathcal{F}$ consists of a single final set, then we say that M is *simple*.

REMARK. A nondeterministic finite automaton is considered to be a special case of an alternating finite automaton whose states are all existential states.

DEFINITION 2.2.2. Let $x = x_0x_1x_2\ldots$ be in Σ^ω , where x_n is in Σ for $n \geq 0$, then a *computation tree* $T(M, x)$ of M on x is an infinite labelled tree satisfying the following conditions:

- (1) $T(M, x)$ has no leaf.
- (2) The nodes are labelled with the elements in S . In particular, the root of $T(M, x)$ is labelled with s_0 .
- (3) For each $n \geq 0$, the edges between level n and level $n+1$ are labelled with x_n .
- (4) If node v in level n is labelled with a universal state s , then v

has a child labelled with q for each q in $\Delta(s, x_n)$.

(5) If node v in level n is labelled with an existential state s , then v has exactly one child labelled with q for some q in $\Delta(s, x_n)$.

REMARK. For nondeterministic finite automata, a computation tree is an infinite path.

DEFINITION 2.2.3. An infinite path α in $T(M, x)$ beginning at the root is called a *run* in $T(M, x)$. For a run α , we define

- (1) $I(\alpha) = \{s \mid \text{state } s \text{ occurs in } \alpha \text{ infinitely many times}\},$
- (2) $O(\alpha) = \{s \mid \text{state } s \text{ occurs in } \alpha\}.$

DEFINITION 2.2.4. For an alternating finite automaton $M = \langle S, f, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ and x in Σ^ω , we say that M *accepts* x in the sense of C_i ($i=1, \dots, 6$), if there exists a computation tree $T(M, x)$ such that for each run α in $T(M, x)$, there exists F in $\mathcal{F}$ satisfying the condition C_i , where

- (C_1) $I(\alpha) \cap F \neq \emptyset,$
- (C_2) $I(\alpha) \subseteq F,$
- (C_3) $O(\alpha) \cap F \neq \emptyset,$
- (C_4) $O(\alpha) \subseteq F,$
- (C_5) $I(\alpha) = F,$
- (C_6) $O(\alpha) = F.$

We call $T(M, x)$ an *accepting computation tree* of M on x in the sense of C_i , respectively. For $i=1, \dots, 6$, we denote by $L_i(M)$ the set of ω -words accepted by M in the sense of C_i and we say that M *recognizes* an ω -language L in the sense of C_i if $L = L_i(M)$.

DEFINITION 2.2.5. For $i=1, \dots, 6$, we define

- (1) $\mathcal{A}_i = \{L_i(M) \mid M \text{ is an afa}\},$
- (2) $\mathcal{A}_i^s = \{L_i(M) \mid M \text{ is a simple afa}\}.$

We summarize in Table 2-1 our new results together with the former results on deterministic and nondeterministic finite automata on ω -words.

As shown in Table 2-1, C_1 type acceptance separates the nondeterministic and deterministic classes and C_2 type acceptance separates the

alternating and nondeterministic classes. On the other hand, no changes occur under C_3 and C_4 type acceptances. For C_5 and C_6 type acceptances of alternating finite automata, we have not yet determined the classes $\mathcal{A}_5$ and $\mathcal{A}_6$ completely. Only partial results are reported in Section 2.5. Therefore we use a symbol U standing for *Unknown* in Table 2-1.

Table 2-1

	C_1	C_2	C_3	C_4	C_5	C_6
$\mathcal{D}_i$	G_θ^R	F_θ^R	G^R	F^R	R	$G_\theta^R \cap F_\theta^R$
$\mathcal{N}_i$	R	F_θ^R	G^R	F^R	R	F_θ^R
$\mathcal{A}_i$	R	R	G^R	F^R	U	U

2.3. Simple alternating finite automata

It is known that a single final set is sufficient for nondeterministic and deterministic cases under acceptance conditions C_i for $i=1, \dots, 4$ (Theorem 1.2.1). The purpose of this section is to show that the same fact holds for alternating finite automata on ω -words.

An alternating finite automaton $M = \langle S, f, \Sigma, \Delta, s_0, \{F\} \rangle$ with a single final set F is also denoted by $M = \langle S, f, \Sigma, \Delta, s_0, F \rangle$.

THEOREM 2.3.1. For $i=1, \dots, 4$, $\mathcal{A}_i = \mathcal{A}_i^s$.

PROOF. Let $M = \langle S, f, \Sigma, \Delta, s_0, \mathcal{F} \rangle$. For each $i=1, \dots, 4$, we construct a simple *afa* $\tilde{M} = \langle \tilde{S}, \tilde{f}, \tilde{M}, \tilde{\Sigma}, \tilde{\Delta}, \tilde{s}_0, \tilde{F} \rangle$ which recognizes the same ω -language as M in the sense of C_i .

Case (i). C_1 : $\tilde{M}$ is simply defined as $\tilde{M} = \langle S, f, \Sigma, \Delta, s_0, \tilde{F} \rangle$, where $\tilde{F} = \{s \mid s \in F \text{ for some } F \text{ in } \mathcal{F}\}$. Let x be in Σ^ω and let $T(M, x)$ be a computation tree of M on x . $T(M, x)$ is also a computation tree of $\tilde{M}$ on x . Then observe that the following two statements are equivalent:

- (1) For each run α in $T(M, x)$, there exists an F in $\mathcal{F}$ such that $I(\alpha) \cap F \neq \emptyset$.
- (2) For each run α in $T(M, x)$, $I(\alpha) \cap \tilde{F} \neq \emptyset$.

Hence M and $\tilde{M}$ recognize the same ω -language in the sense of C_1 .

Case (ii). C_2 : We define a simple *afa* $\tilde{M} = \langle \tilde{S}, \tilde{f}, \tilde{\Sigma}, \tilde{\Delta}, \tilde{s}_0, \tilde{F} \rangle$ in the same way as in [32]:

- (1) $\tilde{S} = \{(s, \mathcal{F}') \mid s \in S, \mathcal{F}' \subseteq \mathcal{F}, s \text{ is in } F \text{ for each } F \text{ in } \mathcal{F}'\}.$
- (2) $\tilde{f}(s, \mathcal{F}') = f(s)$ for $(s, \mathcal{F}')$ in $\tilde{S}$.
- (3) $\tilde{s}_0 = (s_0, \phi).$
- (4) $\tilde{A}((s, \mathcal{F}'), \sigma) = \{(q, \{F \mid F \in \mathcal{F}', q \in F\}) \mid q \in \Delta(s, \sigma)\}$ for all $\phi \neq \mathcal{F}' \subseteq \mathcal{F}$ and $\tilde{A}((s, \phi), \sigma) = \{(q, \{F \mid F \in \mathcal{F}, \text{ and } q \in F\}) \mid q \in \Delta(s, \sigma)\}$
- (5) $\tilde{F} = \{(s, \mathcal{F}') \mid (s, \mathcal{F}') \in \tilde{S} \text{ and } \mathcal{F}' \neq \phi\}.$

Let x be in Σ^ω . Then it should be noticed that there is a one-to-one correspondence between the computation trees of M on x and the computation trees of $\tilde{M}$ on x . Namely, for a computation tree $T(M, x)$ of M on x , we associate it with a computation tree $T(\tilde{M}, x)$ of $\tilde{M}$ on x that begins with the root (s_0, ϕ) and is created by expanding it according to the transitions corresponding to those in $T(M, x)$. Conversely, for $T(\tilde{M}, x)$ we associate it with $T(M, x)$ by replacing label $(s, \mathcal{F}')$ by s for each node. Let $T(M, x)$ and $T(\tilde{M}, x)$ be the computation trees of M on x and $\tilde{M}$ on x corresponding to each other. For a run α in $T(M, x)$ we denote by $\tilde{\alpha}$ the corresponding run in $T(\tilde{M}, x)$ and vice versa. We will show that the following two statements are equivalent:

(6) For each run α in $T(M, x)$, there exists an F in $\mathcal{F}$ such that $I(\alpha) \subseteq F$.

(7) For each run $\tilde{\alpha}$ in $T(\tilde{M}, x)$, $I(\tilde{\alpha}) \subseteq \tilde{F}$.

Let α be

$$s_0 \xrightarrow{x_0} s_1 \xrightarrow{x_1} \dots \xrightarrow{x_{n-1}} s_n \xrightarrow{x_n} \dots$$

and $\tilde{\alpha}$ be

$$(s_0, \mathcal{F}_0) \xrightarrow{x_0} (s_1, \mathcal{F}_1) \xrightarrow{x_1} \dots \xrightarrow{x_{n-1}} (s_n, \mathcal{F}_n) \xrightarrow{x_n} \dots$$

Then note that there exists an integer k_0 such that for any $k \geq k_0$, $I(\alpha) = \{s_n \mid n \geq k\}$. Similarly, there exists an integer $k_1 \geq k_0$ such that $I(\tilde{\alpha}) = \{(s_n, \mathcal{F}_n) \mid n \geq k\}$ for all $k \geq k_1$. Assume that (6) holds. If $\mathcal{F}_m = \phi$ for some $m \geq k_1$, then $\mathcal{F}_n = \phi$ for infinitely many $n \geq k_1$. Since $\mathcal{F}_m = \phi$ and $I(\alpha) = \{s_n \mid n \geq m\} \subseteq F$, we can see by the definition of $\tilde{A}$ that $\mathcal{F}_{m+1}$ contains F . Therefore, again by the definition of $\tilde{A}$, $\mathcal{F}_n$ contains F for all $n > m$. Hence $\mathcal{F}_n \neq \phi$ for all $n > m$. This is a contradiction. Therefore $\mathcal{F}_n \neq \phi$ for all $n \geq k_1$. Hence $I(\tilde{\alpha}) \subseteq \tilde{F}$. Conversely assume that $I(\tilde{\alpha}) \subseteq \tilde{F}$. Then by the definition of $\tilde{A}$, $\mathcal{F}_{n+1} \subseteq \mathcal{F}_n$ and $\mathcal{F}_n \neq \phi$ for all $n \geq k_1$. Since $\mathcal{F}_n$ is finite,

there exists an F such that F is in $\mathcal{F}_n$ for all $n \geq k_1$. Hence s_n is in F for all $n \geq k_1$. Therefore $I(\alpha) \subseteq F$. Hence M and $\tilde{M}$ define the same set of ω -words.

Case (iii). C_3 : The same as Case (i).

Case (iv). C_4 : We define a simple *afa* $\tilde{M} = \langle \tilde{S}, \tilde{f}, \tilde{\Sigma}, \tilde{A}, \tilde{s}_0, \tilde{F} \rangle$ as follows:

- (1) $\tilde{S} = \{(s, \mathcal{F}') \mid s \in S, \mathcal{F}' \subseteq \mathcal{F}, s \text{ is in } F \text{ for each } F \text{ in } \mathcal{F}'\}$.
- (2) $\tilde{f}(s, \mathcal{F}') = f(s)$ for $(s, \mathcal{F}')$ in $\tilde{S}$.
- (3) $\tilde{s}_0 = (s_0, \mathcal{F}_0)$, where $\mathcal{F}_0 = \{F \mid s_0 \in F, F \in \mathcal{F}\}$.
- (4) $\tilde{A}((s, \mathcal{F}'), \sigma) = \{(q, \{F \mid F \in \mathcal{F}', q \in F\}) \mid q \in A(s, \sigma)\}$ for all $\mathcal{F}' \subseteq \mathcal{F}$.
- (5) $\tilde{F} = \{(s, \mathcal{F}') \mid (s, \mathcal{F}') \in \tilde{S} \text{ with } \mathcal{F}' \neq \emptyset\}$.

In the same way as Case (ii), there is a one-to-one correspondence between the computation trees of M on x and the computation trees of $\tilde{M}$ on x . We use the same notations as in Case (ii). We will show the equivalence of the following two statements:

(6) For each run α in $T(M, x)$, there exists an F in $\mathcal{F}$ such that $O(\alpha) \subseteq F$.

(7) For each run $\tilde{\alpha}$ in $T(\tilde{M}, x)$, $O(\tilde{\alpha}) \subseteq \tilde{F}$.

Let α be

$$s_0 \xrightarrow{x_0} s_1 \xrightarrow{x_1} \dots \xrightarrow{x_{n-1}} s_n \xrightarrow{x_n} \dots$$

and $\tilde{\alpha}$ be

$$(s_0, \mathcal{F}_0) \xrightarrow{x_0} (s_1, \mathcal{F}_1) \xrightarrow{x_1} \dots \xrightarrow{x_{n-1}} (s_n, \mathcal{F}_n) \xrightarrow{x_n} \dots$$

Assume that $O(\alpha) = \{s_n \mid n \geq 0\} \subseteq F$ for some F in $\mathcal{F}$. By definition $F \in \mathcal{F}_0$ and therefore $F \in \mathcal{F}_n$ for all $n \geq 0$. Hence $O(\tilde{\alpha}) \subseteq \tilde{F}$. Conversely assume that $O(\tilde{\alpha}) \subseteq \tilde{F}$. Since $\mathcal{F}_{n+1} \subseteq \mathcal{F}_n$ and $\mathcal{F}_n \neq \emptyset$ for all $n \geq 0$, there exists an F such that $F \in \mathcal{F}_n$ for all $n \geq 0$. By the definition of $\tilde{A}$, $s_n \in F$ for all $n \geq 0$. Hence $O(\alpha) \subseteq F$.

2.4. Comparison with the nondeterministic classes

In this section we determine the classes $\mathcal{N}_i$, hence $\mathcal{N}_i^*$, for $i=1, \dots, 4$.

THEOREM 2.4.1. $\mathcal{N}_1 = \mathcal{N}_1^*$.

PROOF¹⁾. The containment $\mathcal{N}_1 \subseteq \mathcal{N}_1^*$ is obvious. Let L be an ω -lan-

1) For notational convenience, levels of a computation tree are changed to begin with 1 instead of 0 in this proof.

guage in $\mathcal{A}_1$ and let $M = \langle S, f, \Sigma, A, s_0, F \rangle$ be a simple *afa* which recognizes L in the sense of C_1 . To show the containment $\mathcal{A}_1 \subseteq \mathcal{A}_1$ we construct a nondeterministic finite automaton M' which recognizes L in the sense of C_1 . Before describing M' we need several preliminary notations and observations. Let x be in Σ^ω and let T be a computation tree of M on x . For a node u of T we denote by $T(u)$ the subtree of T with root u . For $n \geq 1$, $V(n, T)$ denotes the set of the nodes in level n . Given $n \geq m \geq 1$, we divide $V(n, T)$ as follows:

(1) $V_0(n, m, T) = \{u \mid u \text{ is in } V(n, T) \text{ and the path between } u \text{ and the node in level } m \text{ does not hold any element in } F\}$.

(2) $V_1(n, m, T) = V(n, T) - V_0(n, m, T)$.

$V_0(n, m, T) = \emptyset$ means that all the paths between level n and level m hold some elements in F . For $n \geq m \geq 1$ we define an equivalence relation $R(n, m, T)$ on $V(n, T)$ by $uR(n, m, T)v$ if (3) and (4) hold.

(3) Both u and v are in $V_0(n, m, T)$ or both u and v are in $V_1(n, m, T)$.

(4) u and v hold the same label.

Assume that M accepts x in the sense of C_1 . Then we show that there exist an accepting computation tree T of M on x and a sequence $0 = h_0 < h_1 < \dots < h_n < \dots$ of integers satisfying (5) and (6).

(5) $h_{n+1} = \min\{m \mid m > h_n \text{ and } V_0(m, h_n + 1, T) = \emptyset\}$ for $n \geq 0$.

(6) For each n, m with $h_n < m \leq h_{n+1}$, if $uR(m, h_n + 1, T)v$, then the transitions taken at u are the same as those at v .

In order to show that such T and $\{h_n\}_{n=0}^\infty$ exist, we define a sequence $\{T_n\}_{n=0}^\infty$ of accepting computation trees of M on x . We define T_n and h_n inductively. Let $h_0 = 0$ and T_0 be an arbitrary accepting computation tree of M on x . Assume that T_n and h_n are defined but T_{n+1} and h_{n+1} are not. Moreover assume that $n \geq h_n$ and $V_0(l, h_n + 1, T_n) \subseteq V_0(l, h_n + 1, T_{h_n})$ for $l > h_n$. Let $C\{u_1\}, \dots, C\{u_k\} \subseteq V(n+1, T_n)$ be the equivalence classes of $R(n+1, h_n + 1, T_n)$, where $u_j (j=1, \dots, k)$ are their representative nodes in level $n+1$. Then, in T_n , for each $j (j=1, \dots, k)$ we replace the subtree $T_n(u)$ at u by $T_n(u_j)$ for all u in $C\{u_j\}$. Let T_{n+1} be the tree obtained by the above replacement. Then notice that T_{n+1} is an accepting computation tree of M on x . If $V_0(n+1, h_n + 1, T_n) = \emptyset$, then we define $h_{n+1} = n+1$ else

h_{m+1} is not yet defined. Since T_{h_m} is an accepting computation tree, there exists an integer $t > h_m$ such that $V_0(t, h_m+1, T_{h_m}) = \phi$. By (3) and (4) we see that $V_0(l, h_m+1, T_{n+1}) \subseteq V_0(l, h_m+1, T_n)$ for all $l > h_m$. Therefore, $V_0(l, h_m+1, T_{n+1}) \subseteq V_0(l, h_m+1, T_{h_m})$. Since $V_0(t, h_m+1, T_{h_m}) = \phi$, h_{m+1} will be eventually defined. Then note that h_{m+1} is defined to be $\min\{l | l > h_m \text{ and } V_0(l, h_m+1, T_l) = \phi\}$. Hence $\{h_n\}_{n=0}^\infty$ and $\{T_n\}_{n=0}^\infty$ are defined. Then there exists a computation tree T such that T coincides with T_n up to level n for all $n \geq 0$. Then, by the choice of $\{h_n\}_{n=0}^\infty$, $V_0(h_{n+1}, h_n+1, T) = \phi$ for all $n \geq 0$. Hence T is an accepting computation tree of M on x . Obviously, T and $\{h_n\}_{n=0}^\infty$ satisfy (5) and (6).

We now describe the moves of M' . Let $x = x_1 x_2 \dots$ be in Σ^ω . If M accepts x in sense of C_1 , then by the above observation there exist an accepting computation tree T and a sequence $\{h_n\}_{n=0}^\infty$ satisfying (5) and (6). M' has two variables S_0 and S_1 that take subsets of S as values. M' on x simulates T level by level. In simulating the level n ($h_n < n \leq h_{n+1}$) of T , M' keeps the labels of the nodes in $V_i(n, h_m+1, T)$ in S_i for $i=0, 1$. By (3)-(6), M' need not keep all the nodes in level n . Initially, if s_0 is in F , then $S_0 = \phi$ and $S_1 = \{s_0\}$ else $S_0 = \{s_0\}$ and $S_1 = \phi$. Assume that, in level n , $S_0 = S_0^{(n)}$ and $S_1 = S_1^{(n)}$. In level $n+1$, variables S_0 and S_1 are changed in the following way:

(1) $S_0 = S_0^{(n)} \neq \phi$: For each universal state s in $S_0 = S_0^{(n)}$ and for each state q in $\Delta(s, x_n)$, if q is in F , then q is put into S_1 else into S_0 . For each existential state s in $S_0 = S_0^{(n)}$, M' nondeterministically chooses a state q in $\Delta(s, x_n)$ and puts q into S_1 if q is in F else into S_0 . For each universal state s in $S_1 = S_1^{(n)}$ all states in $\Delta(s, x_n)$ are put into S_1 and, for each existential state s in $S_1 = S_1^{(n)}$, exactly one state is nondeterministically chosen from $\Delta(s, x_n)$ and is put into S_1 .

(2) $S_0 = S_0^{(n)} = \phi$: In this case we say that M' is in the reset mode. For each universal state s in $S_1 = S_1^{(n)}$ and for each state q in $\Delta(s, x_n)$, if q is in F , then q is put into S_1 else into S_0 . For each existential state s in $S_1 = S_1^{(n)}$, a state q in $\Delta(s, x_n)$ is nondeterministically chosen and M' puts q into S_1 if q is in F else into S_0 .

Then it should be noticed that M' on x can pass the reset mode infinitely often. Conversely, if M' on x can pass the reset mode infinitely often, then it is easily observed that there is an accepting computation tree of M on x in the sense of C_1 . By this observation, we define the

set F' of final states of M' by the set of states in the reset mode. Then L is recognized by M' in the sense of C_1 .

THEOREM 2.4.2. $\mathcal{A}_3 = \mathcal{N}_3$.

PROOF. For a simple *afa* M , let M' be the same *nfa* as the one constructed in Theorem 2.4.1. Then we can show in a similar way that M' recognizes the same ω -language as M in the sense of C_3 .

THEOREM 2.4.3. $\mathcal{A}_4 = \mathcal{N}_4$.

PROOF. Let $M = \langle S, f, \Sigma, \Delta, s_0, F \rangle$ be a simple *afa*. A simple *nfa* M' that recognizes the same ω -language as M in the sense of C_4 moves as follows: M' has a variable Q which takes a subset of S as a value. Assume that an input $x = x_0 x_1 x_2 \dots$ is given. Initially M' puts $Q = \{s_0\}$. Assume that, at step n , $Q = Q_n$. Q is updated as follows: For each universal state s in Q_n , all states in $\Delta(s, x_n)$ are put into Q and for each existential state s in Q_n , exactly one state in $\Delta(s, x_n)$ is chosen and is put into Q . The set F' of final states of M' is $\{S' \mid \phi \models S' \subseteq F\}$. Then it is easily observed that M accepts x in the sense of C_4 if and only if M' accepts x in the sense of C_4 .

THEOREM 2.4.4. $\mathcal{A}_1 = \mathcal{A}_2$.

PROOF. We show that, for $L \subseteq \Sigma^\omega$, L is in $\mathcal{A}_1$ if and only if $\bar{L} = \Sigma^\omega - L$ is in $\mathcal{A}_2$. We first prove that $L \in \mathcal{A}_1$ implies $\bar{L} \in \mathcal{A}_2$. Assume that L is in $\mathcal{A}_1$ and let $M = \langle S, f, \Sigma, \Delta, s_0, F \rangle$ be a simple *afa* that recognizes L in the sense of C_1 . We show that $\bar{L}$ is in $\mathcal{A}_2$. By definition, the following two statements are equivalent:

- (1) x is in $\bar{L}$.
- (2) For any computation tree $T(M, x)$ there exists a run α in $T(M, x)$ such that $I(\alpha) \subseteq \bar{F}$, where $\bar{F} = S - F$.

Then consider a simple *afa* $\bar{M} = \langle \bar{S}, \bar{f}, \Sigma, \bar{\Delta}, \bar{s}_0, \bar{F} \rangle$, where $\bar{f}(s) = \text{and}$ if $f(s) = \text{or}$, $\bar{f}(s) = \text{or}$ if $f(s) = \text{and}$. Then notice that the following statement is equivalent to (2):

- (3) There exists a computation tree $T(\bar{M}, x)$ of $\bar{M}$ on x such that, for any run α in $T(\bar{M}, x)$, $I(\alpha) \subseteq \bar{F}$.

Hence $\bar{L}$ is in $\mathcal{A}_2$. The converse implication can be proved in the same way. By Theorem 2.4.1 and Theorem 1.3.2, $\mathcal{A}_1$ is closed under complement. Therefore by the above equivalence we see that $\mathcal{A}_1 = \mathcal{A}_2$.

2.5. Concluding remarks

For the classes of ω -languages defined by alternating finite automata under C_5 and C_6 type acceptance conditions we have the following results.

PROPOSITION 2.5.1. $R \subseteq \mathcal{A}_5$.

PROOF. Given a simple *afa* $M = \langle S, f, \Sigma, \Delta, s_0, F \rangle$, we define an *afa* $M' = \langle S, f, \Sigma, \Delta, s_0, \mathcal{F} \rangle$, where $\mathcal{F} = \{G \mid G \cap F \neq \emptyset, G \subseteq S\}$. Then we can see that $L_1(M) = L_5(M')$. Thus we have $\mathcal{A}_1 = R \subseteq \mathcal{A}_5$.

PROPOSITION 2.5.2. $F_\sigma^R \subseteq \mathcal{A}_6 \subseteq R$.

PROOF. By Table 2-1, $F_\sigma^R = \mathcal{A}_6 \subseteq \mathcal{A}_6$ for the first inclusion.

For the second inclusion, given an *afa* $M = \langle S, f, \Sigma, \Delta, s_0, \mathcal{F} \rangle$, we construct a simple *afa* $M' = \langle S', f', \Sigma, \Delta', s'_0, F' \rangle$ such that $L_6(M) = L_1(M')$ as follows:

- (1) $S' = S \times P(S)$.
- (2) $s'_0 = (s_0, \{s_0\})$.
- (3) $f'(s, Q) = f(s)$ for (s, Q) in S' .
- (4) For each $s \in S$, $Q \subseteq S$ and $\sigma \in \Sigma$, $\Delta'((s, Q), \sigma) = \{(q, Q_1) \mid q \in \Delta(s, \sigma), Q_1 = Q \cup \{q\}\}$.
- (5) $F' = \{(s, F) \mid s \in S, F \in \mathcal{F}\}$.

As simulating M , M' stores the states occurring so far in the second component of its state. Let $x = x_0 x_1 \dots$ be in Σ^ω . Then it should be noticed that there is a one-to-one correspondence between the computation trees of M on x and the computation trees of M' on x . Namely, for a computation tree $T(M, x)$ of M on x , we associate it with a computation tree $T(M', x)$ of M' on x that begins with the root $(s_0, \{s_0\})$ and is created by expanding it according to the transitions corresponding to those in $T(M, x)$. Conversely, for $T(M', x)$ we associate it with $T(M, x)$ by replacing label (s, Q) by s for each node. Let $T(M, x)$ and $T(M', x)$ be the computation trees of M on x and M' on x corresponding to each

other. For a run α in $T(M, x)$ we denote by α' the corresponding run in $T(M', x)$ and vice versa. We will show that the following two statements are equivalent:

(6) For each run α in $T(M, x)$, there exists an F in $\mathcal{F}$ such that $O(\alpha) = F$.

(7) For each run α' in $T(M', x)$, $I(\alpha') \cap F' \neq \emptyset$.

Let α be

$$s_0 \xrightarrow{x_0} s_1 \xrightarrow{x_1} \dots \xrightarrow{x_{n-1}} s_n \xrightarrow{x_n} \dots$$

and α' be

$$(s_0, F_0) \xrightarrow{x_0} (s_1, F_1) \xrightarrow{x_1} \dots \xrightarrow{x_{n-1}} (s_n, F_n) \xrightarrow{x_n} \dots$$

Then note that $F_n = \{s_i \mid 0 \leq i \leq n\}$ and $F_n \subseteq F_{n+1}$ for each $n \geq 0$ by (4). Assume that (6) holds. Then there exists an integer n_0 such that $F = F_n$ for all $n \geq n_0$. Since $\{s_n \mid n \geq n_0\}$ is finite, there exists s in S such that $(s, F) = (s_n, F_n)$ for infinitely many $n \geq n_0$. Therefore $I(\alpha') \cap F' \neq \emptyset$. Conversely assume that $I(\alpha') \cap F' \neq \emptyset$. Since $F_n \subseteq F_{n+1}$ for $n \geq 0$, there exists an integer n_0 such that $F_{n_0} = F_n$ for all $n \geq n_0$. Note that $F_{n_0} = \{s_n \mid n \geq 0\}$. Because $I(\alpha') \cap F' \neq \emptyset$, there is (s, F) in F' such that $(s, F) = (s_n, F_n)$ for infinitely many $n \geq 0$. Since $F_{n_0} = F_n$ for all $n \geq n_0$, $F = F_{n_0}$. Therefore $F_{n_0} = \{s_n \mid n \geq 0\} = O(\alpha) = F$ in $\mathcal{F}$. Hence M and M' define the same set of ω -words.

Thus $\mathcal{A}'_6 \subseteq \mathcal{A}'_1 = \mathcal{N}'_1 = R$ by Theorem 2.4.1.

In the definition of alternating finite automata we have assumed that $\Delta(s, \sigma) \neq \emptyset$ for each s in S and each σ in Σ . We call an *afa* which allows the case $\Delta(s, \sigma) = \emptyset$ a *partial afa*. Let $\mathcal{P}\mathcal{A}_i$ ($i=1, \dots, 6$) be the classes of ω -languages recognized by partial *afa*'s in the sense of C_i . It is not hard to show that the partial and nonpartial classes are equal under all but C_3 type acceptances. This can be shown by constructing a nonpartial alternating finite automaton by adding a trapping state into which all undefined transitions come. For C_3 type acceptance, we can prove that $\mathcal{P}\mathcal{A}_3 = \mathcal{P}\mathcal{N}_3$ by taking care of the undefined transitions in the construction of the *nfa* in the proof of Theorem 2.4.1.

As combined with the results in Section 1.5, Table 2-2 summarizes the classes for partial finite automata on ω -words.

Table 2-2

	C_1	C_2	C_3	C_4	C_5	C_6
$\mathcal{PD}_i$	G_σ^R	F_σ^R	$G^R \Delta F^R$	F^R	R	$G_\sigma^R \cap F_\sigma^R$
$\mathcal{PN}_i$	R	F_σ^R	F_σ^R	F^R	R	F_σ^R
$\mathcal{PA}_i$	R	R	F_σ^R	F^R	$\mathcal{A}_5$	$\mathcal{A}_6$

Chapter 3. Finite tree automata on infinite trees

3.1. Introduction

Rabin [27–29] introduced finite tree automata on infinite trees to study the problems of definability in monadic second-order theories. As by-products, he obtained several decidability results. He considered the acceptance conditions which are extensions of C_1 and C_5 types discussed in Chapters 1 and 2 to infinite tree cases. In this chapter we deal with all six types of acceptance conditions extended for finite tree automata on infinite trees which move from the root infinitely, and study the classes of infinite tree languages defined by finite tree automata under these six types of acceptances. Each condition uses a family of final sets for acceptance as was done in Chapters 1 and 2 for finite automata on ω -words. The purpose of this chapter is to establish the relationship among these infinite tree language classes. First it is proved that a single final set is sufficient for four types of acceptance definitions. We also prove an embedding theorem which relates the infinite tree language classes with the ω -language classes. With the aid of this embedding theorem and a result due to Rabin [28], we classify the infinite tree language classes. A summary of the results is figured out in Fig. 3-1 in Section 3.5.

It is known that the nondeterministic infinite tree language class is larger than the deterministic one for any type of the acceptances [30]. To refine this result, we consider the nondeterministic degree which is defined to be the maximum number of possible transitions of a finite tree automaton. Then the determinism is regarded as the nondeterministic degree one. We show that the nondeterministic degree gives a hierarchy between the nondeterministic class and the deterministic one for each acceptance definition. This is a property special to (infinite) tree languages. In addition to this observation, the results in this chap-

ter show that the class of infinite tree languages accepted by nondeterministic finite tree automata which accept in the sense of Muller (C_s type acceptance) is the most general one and this class may be considered as a natural extension of the class of ω -regular languages [21], where deterministic finite automata are used as basis.

3.2. Preliminaries

Let $\Sigma_k = \{0, 1, \dots, k-1\}$ be an alphabet consisting of k symbols. The *infinite k -ary tree* is the set $T_k = \Sigma_k^*$. The empty string ϵ is the root of T_k . For $x \in T_k$, $x0, x1, \dots, x(k-1)$ are the sons of x arranged from left to right. Note that $T_k \subset T_{k+1}$ for $k \geq 1$. Let Σ be a finite alphabet. An *infinite k -ary tree over Σ* (a *k -ary Σ -tree*) is a function $t: T_k \rightarrow \Sigma$. The set of all k -ary Σ -trees is denoted by Σ^T_k . For x, y in T_k , we say that x is a prefix of y denoted by $x \leq y$ if $y = xz$ for some z in Σ_k^* . The prefix relation gives a partial ordering on T_k . A *path* in T_k is a subset $\pi \subseteq T_k$ satisfying (i) $\epsilon \in \pi$, (ii) for $x \in \pi$, exactly one of k sons $x0, x1, \dots, x(k-1)$ belongs to π , (iii) π is the smallest subset of T_k satisfying (i) and (ii).

For a function $f: X \rightarrow Y$ and a subset X' of X , $f|X'$ denotes the restriction of f to X' .

DEFINITION 3.2.1. A (*nondeterministic*) *finite k -ary tree automaton* is a quintuple $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$, where

- (1) S is a finite set of states,
- (2) Δ is a *transition function* $\Delta: S \times \Sigma \rightarrow P(S^k) - \{\emptyset\}$, where S^k is the k -fold product of S ,
- (3) s_0 is a state in S called the *initial state*,
- (4) $\mathcal{F}$ is a family of subsets of S . F in $\mathcal{F}$ is called a *final set* and states in F are called *final states*. If $\mathcal{F}$ consists of a single final set, then we say that M is *simple*.

M is said to be *deterministic* if $|\Delta(s, \sigma)| = 1$ for each s in S and each σ in Σ .

Let $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ be a finite k -ary tree automaton and let $t: T_k \rightarrow \Sigma$ be a k -ary Σ -tree. A *run* of M on t is a function $r: T_k \rightarrow S$ such that $r(\epsilon) = s_0$ and for each x in T_k , $(r(x0), r(x1), \dots, r(x(k-1))) \in \Delta(r(x), t(x))$. Let π be a path in T_k . For a run $r: T_k \rightarrow S$ we define

$$I(r|\pi) = \{s | s \in S, s = r(x) \text{ for infinitely many } x \text{ in } \pi\}.$$

DEFINITION 3.2.2. A finite k -ary tree automaton $M = \langle S, \Sigma, \delta, s_0, \mathcal{F} \rangle$ is said to *accept* a k -ary Σ -tree t in the sense of $C_i (i=1, \dots, 6)$ if there exists a run r of M on t such that for each path $\pi \subseteq T_k$ there is a final set F in $\mathcal{F}$ which satisfies the condition C_i described below. A run r satisfying the condition C_i is called a C_i -*accepting run* of M on t .

$$(C_1) \quad I(r|\pi) \cap F \neq \emptyset.$$

$$(C_2) \quad I(r|\pi) \subseteq F.$$

$$(C_3) \quad r(\pi) \cap F \neq \emptyset.$$

$$(C_4) \quad r(\pi) \subseteq F.$$

$$(C_5) \quad I(r|\pi) = F.$$

$$(C_6) \quad r(\pi) = F.$$

DEFINITION 3.2.3. For $i=1, \dots, 6$, the set $L_i(M)$ of k -ary Σ -trees is defined as

$$L_i(M) = \{t \mid t \in \Sigma^{T_k}, t \text{ is accepted by } M \text{ in the sense of } C_i\}.$$

We say that a set $L \subseteq \Sigma^{T_k}$ is *accepted* by M in the sense of C_i if $L = L_i(M)$.

Definition 3.2.4. For $i=1, \dots, 6$, we define

(1) $\mathcal{N}_i^{(k)} = \{L_i(M) \mid M \text{ is a nondeterministic finite } k\text{-ary tree automaton}\},$

(2) $\mathcal{D}_i^{(k)} = \{L_i(M) \mid M \text{ is a deterministic finite } k\text{-ary tree automaton}\}.$

Here we are mainly concerned with the classes for $k=1, 2$. We denote $\mathcal{N}_i^{(1)}$ (resp. $\mathcal{D}_i^{(1)}$) by $\mathcal{N}_i$ (resp. $\mathcal{D}_i$). The classes $\mathcal{N}_i^{(2)}$ and $\mathcal{D}_i^{(2)}$ are specially denoted by $\mathcal{N}_{\mathcal{T}_i}$ and $\mathcal{D}_{\mathcal{T}_i}$, respectively. A binary Σ -tree is simply called a Σ -tree and a finite binary tree automaton is called a finite tree automaton. For $k=1$, this is just the case of finite automata on ω -words treated in Chapter 1. In fact, $T_1 = \{0\}^*$ and Σ^{T_1} is the same as the set Σ^ω of ω -words. We identify a finite unary tree automaton with a finite automaton.

3.3. Simple finite tree automata

In this section we consider finite tree automata which have just one final set. As was shown in Chapters 1 and 2, a single final set is sufficient for the ω -word acceptances by C_i conditions for $i=1, \dots, 4$. The purpose of this section is to show that an analog also holds for infinite tree acceptances.

A simple finite tree automaton $M = \langle S, \Sigma, \Delta, s_0, \{F\} \rangle$ with a final set F is also denoted by $M = \langle S, \Sigma, \Delta, s_0, F \rangle$.

THEOREM 3.3.1. *Let $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ be a finite tree automaton. Then for each $i=1, \dots, 4$, there exists a simple finite tree automaton $\tilde{M} = \langle \tilde{S}, \tilde{\Sigma}, \tilde{\Delta}, \tilde{s}_0, \tilde{F} \rangle$ such that $L_i(M) = L_i(\tilde{M})$. Furthermore, if M is deterministic, so is $\tilde{M}$.*

PROOF. The idea of the proof is due to [32, Lemma 7].

Case 1. $i=1, 3$: $\tilde{M}$ is simply defined by $\tilde{M} = \langle S, \Sigma, \Delta, s_0, \tilde{F} \rangle$, where $\tilde{F} = \{s \mid s \in F \text{ for some } F \text{ in } \mathcal{F}\}$.

Case 2. $i=2$: $\tilde{M}$ is defined as follows:

- (1) $\tilde{S} = \{(s, \tilde{\mathcal{F}}) \mid s \in S, \tilde{\mathcal{F}} \subseteq \mathcal{F}, \text{ and for each } F \text{ in } \tilde{\mathcal{F}}, s \text{ is in } F\}$.
- (2) $\tilde{s}_0 = (s_0, \phi)$.
- (3) If (s_1, s_2) is in $\Delta(s, \sigma)$, then for all $\phi \not\equiv \tilde{\mathcal{F}} \subseteq \mathcal{F}$, $\tilde{\Delta}((s, \tilde{\mathcal{F}}), \sigma)$ contains $((s_1, \{F \mid F \in \tilde{\mathcal{F}}, s_1 \in F\}), (s_2, \{F \mid F \in \tilde{\mathcal{F}}, s_2 \in F\}))$ and $\tilde{\Delta}((s, \phi), \sigma)$ contains $((s_1, \{F \mid F \in \mathcal{F}, s_1 \in F\}), (s_2, \{F \mid F \in \mathcal{F}, s_2 \in F\}))$.
- (4) $\tilde{F} = \{(s, \tilde{\mathcal{F}}) \mid (s, \tilde{\mathcal{F}}) \in \tilde{S} \text{ with } \tilde{\mathcal{F}} \not\equiv \phi\}$.

Let $t: T_2 \rightarrow \Sigma$ be a Σ -tree. First we establish a one-to-one correspondence between the runs of M on t and the runs of $\tilde{M}$ on t . Let $\tilde{r}: T_2 \rightarrow \tilde{S}$ be an arbitrary run of $\tilde{M}$ on t . Then we associate it with a run $r: T_2 \rightarrow S$ such that $r(x) = s$, where $\tilde{r}(x) = (s, \tilde{\mathcal{F}})$ for x in T_2 . By (3), we can see that r is a run of M on t . Conversely for a run $r: T_2 \rightarrow S$ of M on t , we associate it with a run $\tilde{r}: T_2 \rightarrow \tilde{S}$ of $\tilde{M}$ on t which is defined as follows: $\tilde{r}$ is inductively defined. First we define $\tilde{r}(\epsilon) = (s_0, \phi)$. Suppose that $\tilde{r}(x) = (s, \tilde{\mathcal{F}})$ is defined. Then note that $r(x) = s$. Assume that in the run r a transition $(s_1, s_2) \in \Delta(s, t(x))$ is taken at the node x . Then if $\tilde{\mathcal{F}} = \phi$, the transition $((s_1, \{F \mid F \in \mathcal{F}, s_1 \in F\}), (s_2, \{F \mid F \in \mathcal{F}, s_2 \in F\}))$ is taken at x in the run $\tilde{r}$. Namely we define $\tilde{r}(x0) = (s_1, \{F \mid F \in \mathcal{F}, s_1 \in F\})$ and $\tilde{r}(x1) = (s_2, \{F \mid F \in \mathcal{F}, s_2 \in F\})$. If $\tilde{\mathcal{F}} \not\equiv \phi$, then we put $\tilde{r}(x0) = (s_1, \{F \mid F \in \tilde{\mathcal{F}}, s_1 \in F\})$ and $\tilde{r}(x1) = (s_2, \{F \mid F \in \tilde{\mathcal{F}}, s_2 \in F\})$. Again by (3) we see that $\tilde{r}$ is a run of $\tilde{M}$ on t . By this correspondence we assume that r and $\tilde{r}$ denote the runs corresponding to each other. To prove $L_2(M) = L_2(\tilde{M})$, it suffices to show that the following two statements are equivalent for each path $\pi \subseteq T_2$.

- (5) There exists F in $\mathcal{F}$ such that $I(r|\pi) \subseteq F$.
- (6) $I(\tilde{r}|\pi) \subseteq \tilde{F}$.

For a path $\pi = \{x_0 < x_1 < \dots x_n < \dots\}$, let $\tilde{r}(x_n) = (s_n, \mathcal{F}_n)$ for $n \geq 0$. Then there exists an integer n_0 such that $I(r|\pi) = \{s_n | n \geq n_0\}$ and $I(\tilde{r}|\pi) = \{(s_n, \mathcal{F}_n) | n \geq n_0\}$. Suppose that $I(r|\pi) \subseteq F$ for some F in $\mathcal{F}$. Then by the choice of n_0 we see that s_n is in F for all $n \geq n_0$. If $\mathcal{F}_{n_1} = \phi$ for some $n_1 \geq n_0$, then by the choice of n_0 we see that $\mathcal{F}_n = \phi$ for infinitely many $n \geq n_1$. As we observed above, s_{n_1+1} is in F . Therefore by (3) we see that F is in $\mathcal{F}_{n_1+1}$. Moreover, since s_n is in F for all $n \geq n_1$, it follows from (3) that F is in $\mathcal{F}_n$ for all $n > n_1$. Hence $\mathcal{F}_n \neq \phi$ for all $n > n_1$. This is a contradiction. Thus $\mathcal{F}_n \neq \phi$ for all $n \geq n_0$. Therefore $I(\tilde{r}|\pi) \subseteq \tilde{F}$. Conversely suppose that $I(\tilde{r}|\pi) \subseteq \tilde{F}$. Then it follows from (3) and (4) that $\mathcal{F}_n \supseteq \mathcal{F}_{n+1}$ and $\mathcal{F}_n \neq \phi$ for all $n \geq n_0$. Since $\mathcal{F}_n$ is finite and $\mathcal{F}_n \neq \phi$, there exists F in $\mathcal{F}$ such that F is in $\mathcal{F}_n$ for all $n \geq n_0$. Therefore s_n is in F for all $n \geq n_0$. Thus $I(r|\pi) \subseteq F$.

Case 3. $i=4$: $\tilde{M}$ is defined as follows:

- (1) $\tilde{S} = \{(s, \tilde{\mathcal{F}}) | s \in S, \tilde{\mathcal{F}} \subseteq \mathcal{F}, \text{ and for each } F \text{ in } \tilde{\mathcal{F}}, s \text{ is in } F\}$.
- (2) $\tilde{s}_0 = (s_0, \mathcal{F}_0)$, where $\mathcal{F}_0 = \{F | F \in \mathcal{F}, s_0 \in F\}$.
- (3) If (s_1, s_2) is in $\mathcal{A}(s, \sigma)$, then for all $\tilde{\mathcal{F}} \subseteq \mathcal{F}$, $\tilde{\mathcal{A}}((s, \tilde{\mathcal{F}}), \sigma)$ contains $((s_1, \{F | F \in \tilde{\mathcal{F}}, s_1 \in F\}), (s_2, \{F | F \in \tilde{\mathcal{F}}, s_2 \in F\}))$.
- (4) $\tilde{F} = \{(s, \tilde{\mathcal{F}}) | (s, \tilde{\mathcal{F}}) \in \tilde{S} \text{ with } \tilde{\mathcal{F}} \neq \phi\}$.

Let $t: T_2 \rightarrow \Sigma$ be a Σ -tree. As in Case 2, by the one-to-one correspondence between the runs of M on t and the runs of $\tilde{M}$ on t , we denote by r and $\tilde{r}$ the runs corresponding to each other. Let $\pi = \{x_0 < x_1 < \dots x_n < \dots\}$ be an arbitrary path in T_2 and let $\tilde{r}(x_n) = (s_n, \mathcal{F}_n)$ for $n \geq 0$. Then we show the equivalence of the following two statements.

- (5) There exists F in $\mathcal{F}$ such that $r(\pi) \subseteq F$.
- (6) $\tilde{r}(\pi) \subseteq \tilde{F}$.

Suppose that $r(\pi) \subseteq F$ in some F in $\mathcal{F}$. Since $r(\varepsilon) = s_0$ is in F , we see that F is in $\mathcal{F}_0$. By induction we can see that F is in $\mathcal{F}_n$ for all $n \geq 0$. Hence $\mathcal{F}_n \neq \phi$ for all $n \geq 0$. Therefore $\tilde{r}(\pi) \subseteq \tilde{F}$. Conversely, if $\tilde{r}(\pi) \subseteq \tilde{F}$, then $\mathcal{F}_n \neq \phi$ for all $n \geq 0$. Since $\mathcal{F}_n \supseteq \mathcal{F}_{n+1}$ and $\mathcal{F}_n \neq \phi$ for $n \geq 0$, there exists F such that F is in $\mathcal{F}_n$ for all $n \geq 0$. Since $\tilde{r}(x_n) = (r(x_n), \mathcal{F}_n)$, it follows from (3) that $r(x_n)$ is in F for all $n \geq 0$. Hence $r(\pi) \subseteq F$.

In Cases 1-3, the construction of $\tilde{M}$ shows that if M is deterministic then so is $\tilde{M}$.

By Theorem 3.3.1, we hereafter consider only simple finite tree automata for C_i -acceptances for $i=1, \dots, 4$.

For the acceptance conditions C_5 and C_6 , the number of final sets affects the definability. We will discuss this matter in Section 3.6.

3.4. Embedding theorem

Let $\Sigma = \{a, b\}$. We define $\iota: \Sigma^{T_1} \rightarrow \Sigma^{T_2}$ and $\rho: \Sigma^{T_2} \rightarrow \Sigma^{T_1}$ as follows: For w in Σ^{T_1} , $\iota(w)$ is defined by $\iota(w)(x) = w(x)$ for x in T_1 and $\iota(w)(x) = a$ for x in $T_2 - T_1$. For t in Σ^{T_2} , we define $\rho(t) = t|T_1$. Namely, $\iota(w)$ is a Σ -tree whose leftmost path is the same as w and other nodes are labelled with a . $\rho(t)$ is the ω -word obtained by spelling out the labels of the leftmost path of t .

THEOREM 3.4.1. *For $i=1, \dots, 6$, if L is in $\mathcal{NST}_i$ (resp. $\mathcal{DT}_i$), then $\rho(L)$ is in $\mathcal{N}_i$ (resp. $\mathcal{D}_i$).*

PROOF. Case 1. $i=1$: Given a finite tree automaton $M = \langle S, \Sigma, \Delta, s_0, F \rangle$ with $L = L_1(M)$, we construct a finite automaton $\tilde{M} = \langle \tilde{S}, \Sigma, \tilde{\Delta}, \tilde{s}_0, \tilde{F} \rangle$ as follows: For s in S , let $M_s = \langle S, \Sigma, \Delta, s, F \rangle$ and let $S_1 = \{s \in S \mid L_1(M_s) \neq \emptyset\}$. Then we define as follows:

(1) $\tilde{S} = S \cup \{\perp\}$, where $\perp \notin S$.

(2) $\tilde{s}_0 = s_0$

(3) For each s in S and σ in Σ , $\tilde{\Delta}(s, \sigma)$ is equal to the set $\{p \mid \text{there is } (p, q) \in \Delta(s, \sigma) \text{ with } q \in S_1\}$ if this set is not empty else $\{\perp\}$. And $\tilde{\Delta}(\perp, \sigma) = \{\perp\}$.

(4) $\tilde{F} = F$.

It t is in L , then there exists a run r of M on t such that for any path π in T_2 , $I(r|_\pi) \cap F \neq \emptyset$. Therefore $r(0^n 1)$ is in S_1 for all $n \geq 0$. Hence $\tilde{r}$ defined by $\tilde{r}(x) = r(x)$ for x in T_1 is a run of $\tilde{M}$ on $\rho(t)$. Moreover $\tilde{r}$ is a C_1 -accepting run since $I(\tilde{r}|_{\pi_0}) \cap F \neq \emptyset$ for the leftmost path π_0 in T_2 . Thus $\rho(L) \subseteq L_1(\tilde{M})$. Conversely if w is in $L_1(\tilde{M})$, then it follows from the definitions of S_1 and $\tilde{\Delta}$ that there exist a Σ -tree t and a C_1 -accepting run of M on t such that $w = \rho(t)$. Therefore $L_1(\tilde{M}) \subseteq \rho(L)$. Thus $\rho(L) = L_1(\tilde{M})$.

Case 2. $i=2$: This case can be proved in the same way as Case 1.

Case 3. $i=3$: Let $M = \langle S, \Sigma, \Delta, s_0, F \rangle$ be a finite tree automaton with $L = L_3(M)$. Without loss of generality, we may assume that s_0 is not in F . We construct a finite automaton $\tilde{M} = \langle \tilde{S}, \Sigma, \tilde{\Delta}, \tilde{s}_0, \tilde{F} \rangle$ satisfying $\rho(L) = L_3(\tilde{M})$ in the following way: Let $S_3 = \{s \in S \mid L_3(M_s) \neq \emptyset\}$. Notice that if s is in F then $L_3(M_s) = \Sigma^{*2}$. Hence $F \subseteq S_3$. For each s in S and σ in Σ , let $S_{(s, \sigma)} = \{p \mid \text{there is } (p, q) \in \Delta(s, \sigma) \text{ with } q \in S_3\}$.

- (1) $S = \tilde{S} \cup \{\perp, \top\}$, where $\perp, \top \notin S$.
- (2) $\tilde{s}_0 = s_0$.
- (3) For each s in S and σ in Σ , $\tilde{\Delta}(s, \sigma)$ is defined by

$$\tilde{\Delta}(s, \sigma) = \begin{cases} \{\perp\} & \text{if } S_{(s, \sigma)} = \emptyset \\ \{\top\} & \text{if } S_{(s, \sigma)} \cap F \neq \emptyset \\ S_{(s, \sigma)} & \text{otherwise} \end{cases}$$

and $\tilde{\Delta}(\perp, \sigma) = \{\perp\}$, $\Delta(\top, \sigma) = \{\top\}$.

- (4) $\tilde{F} = \{\top\}$.

Let t be in L and let r be a C_3 -accepting run of M on t . Then we inductively define $\tilde{r}: T_1 \rightarrow \tilde{S}$ as follows: Let $\tilde{r}(\varepsilon) = r(\varepsilon) = s_0$. Assume that $\tilde{r}(0^n)$ is defined. If $r(0^n)$ is in S , then two cases arise. First if $r(0^{n+1})$ is in F , then $\tilde{r}(0^{n+1}) = \top$. If $r(0^{n+1})$ is $S-F$, then $\tilde{r}(0^{n+1}) = r(0^{n+1})$. If $\tilde{r}(0^n) = \top$, then $\tilde{r}(0^{n+1}) = \top$. Since $r(\pi_0) \cap F \neq \emptyset$ for the leftmost path π_0 , there exists an integer $n_0 \geq 0$ such that $\tilde{r}(0^n)$ is in $S-F$ for $0 \leq n \leq n_0$ and $\tilde{r}(0^n) = \top$ for all $n > n_0$. Since r is a C_3 -accepting run of M on t and since $r(0^n)$ is in $S-F$ for $0 \leq n \leq n_0$, $r(0^{n+1}) \in S_{(r(0^n), t(0^n))}$ for $0 \leq n \leq n_0$ and $r(0^{n_0+1}) \in S_{(r(0^{n_0}), t(0^{n_0}))} \cap F$, where $m = n_0$. Therefore $\tilde{r}$ is a C_3 -accepting run of $\tilde{M}$ on $\rho(t)$. Hence $\rho(L) \subseteq L_3(\tilde{M})$. Conversely, let $\tilde{r}$ be a C_3 -accepting run of $\tilde{M}$ on w in $L_3(\tilde{M})$. Then by (3) and (4), there exists an integer $n_0 \geq 0$ such that $\tilde{r}(0^n)$ is in $S-F$ for $0 \leq n \leq n_0$ and $\tilde{r}(0^n) = \top$ for all $n > n_0$. Furthermore for each n , $0 \leq n \leq n_0$, there is q_n in S_3 such that $(\tilde{r}(0^{n+1}), q_n) \in \Delta(\tilde{r}(0^n), w(0^n))$ and $\tilde{r}(0^{n_0+1}) \in F \subseteq S_3$. Therefore by the definition of S_3 , we can see that there exist a Σ -tree t with $\rho(t) = w$ and a C_3 -accepting run of M on t .

Case 4. $i=4$: This case can be proved in a manner similar to Case 1.

Case 5. $i=5$: For a finite tree automaton $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ with $L = L_5(M)$, we construct a finite automaton $\tilde{M} = \langle \tilde{S}, \Sigma, \tilde{\Delta}, \tilde{s}_0, \tilde{\mathcal{F}} \rangle$ as follows: Let $S_5 = \{s \in S \mid L_5(M_s) \neq \emptyset\}$. Then we define as follows:

- (1) $\tilde{S} = S \cup \{\perp\}$, where $\perp \notin S$.

$$(2) \quad \tilde{s}_0 = s_0.$$

(3) For each s in S and σ in Σ , $\tilde{A}(s, \sigma)$ is equal to the set $\{p \mid \text{there is } (p, q) \in A(s, \sigma) \text{ with } q \in S_s\}$ if this set is not empty else $\{\perp\}$. And $\tilde{A}(\perp, \sigma) = \{\perp\}$.

$$(4) \quad \tilde{\mathcal{F}} = \mathcal{F}.$$

Then by an argument similar to Case 1, we can prove $\rho(L) = L_6(\tilde{M})$.

Case 6. $i=6$: Let $M = \langle S, \Sigma, A, s_0, \mathcal{F} \rangle$ be a finite tree automaton satisfying $L = L_6(M)$. For $\phi \not\equiv Q \subseteq S$ and s in S , let $\mathcal{F}_Q = \{F \subseteq S \mid F \cup Q \in \mathcal{F}\}$ and let $M_{(Q,s)} = \langle S, \Sigma, A, s, \mathcal{F}_Q \rangle$. Then let $S_Q = \{s \in Q \mid L_6(M_{(Q,s)}) \not\equiv \phi\}$. Without loss of generality, we may assume that every F in $\mathcal{F}$ contains the initial state s_0 . For F in $\mathcal{F}$, an F -chain is a set $\{S_i \mid 0 \leq i \leq n\}$ of subsets of F such that $S_0 = \{s_0\}$, $S_n = F$, $S_i \subseteq S_{i+1}$ and $|S_i| = i+1$ for $0 \leq i \leq n$. Then let $\mathcal{F}_F$ be the set of all F -chains. We define a finite automaton $\tilde{M} = \langle \tilde{S}, \Sigma, \tilde{A}, \tilde{s}_0, \tilde{\mathcal{F}} \rangle$ as follows:

$$(1) \quad \tilde{S} = S \times P(S) \cup \{\perp\}$$

$$(2) \quad \tilde{s}_0 = (s_0, \{s_0\}).$$

(3) For each s in S , σ in Σ and $Q \subseteq S$, $\tilde{A}((s, Q), \sigma)$ is equal to the set $\{(p, Q \cup \{p\}) \mid \text{there is } (p, q) \in A(s, \sigma) \text{ with } q \in S_{Q \cup \{q\}}\}$ if this set is not empty else $\{\perp\}$. And $\tilde{A}(\perp, \sigma) = \{\perp\}$.

(4) $\tilde{\mathcal{F}} = \{G \subseteq S \times P(S) \mid p_1(G) = F \text{ and } p_2(G) \in \mathcal{F}_F \text{ for some } F \text{ in } \mathcal{F}\}$, where p_1 and p_2 are the projections to the first and second components, respectively.

For t in $L_6(M)$, let r be a C_6 -accepting run of M on t . Let $s_n = r(0^n)$ and $F_n = \{s_i \mid 0 \leq i \leq n\}$ for $n \geq 0$. Then we define $\tilde{r}: T_1 \rightarrow \tilde{S}$ by $\tilde{r}(0^n) = (s_n, F_n)$ for $n \geq 0$. We show that $\tilde{r}$ is a C_6 -accepting run of $\tilde{M}$ on $\rho(t)$. First note that $\tilde{r}(\epsilon) = (s_0, \{s_0\})$. For each $n \geq 0$, consider the transition $(r(0^{n+1}), r(0^n 1)) \in A(r(0^n), t(0^n))$. Since r is a C_6 -accepting run of M on t , $r(0^n 1)$ is in $S_{F_n \cup \{r(0^n 1)\}}$. Therefore $\tilde{r}$ is a run of $\tilde{M}$ on $\rho(t)$. Since $r(\pi_0)$ is in $\mathcal{F}$ for the leftmost path π_0 in T_2 , $F = \{s_n \mid n \geq 0\}$ is in $\mathcal{F}$. Since $F_0 = \{s_0\}$, $F_n \subseteq F_{n+1}$ for all $n \geq 0$ and $F = \bigcup_{n=0}^{\infty} F_n$, the set $\{(s_n, F_n) \mid n \geq 0\}$ is in $\tilde{\mathcal{F}}$. Hence $\tilde{r}$ is a C_6 -accepting run of $\tilde{M}$ on $\rho(t)$. Thus $\rho(L) \subseteq L_6(\tilde{M})$. Conversely, for a C_6 -accepting run $\tilde{r}$ of $\tilde{M}$ on w in $L_6(\tilde{M})$, we can find a Σ -tree t with $\rho(t) = w$ and a C_6 -accepting run of M on t by using the definitions of S_Q and $\tilde{A}$. Thus $L_6(\tilde{M}) \subseteq \rho(L)$.

In Cases 1-6, it is obvious that if M is deterministic then so is $\tilde{M}$.

REMARK. For a finite tree automaton $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$, we define a finite automaton $M' = \langle S, \Sigma, \Delta', s_0, \mathcal{F} \rangle$ by setting $\Delta'(s, \sigma) = \{p \mid (p, q) \in \Delta(s, \sigma)\}$ for s in S and σ in Σ . Then M' does not necessarily accept $\rho(L_i(M))$ in the sense of C_i for $i=1, \dots, 6$. Namely, the definitions of $\tilde{M}$ in the proof of Theorem 3.4.1 are essential. For example, let $M_i = \langle S, \Sigma, \Delta_i, s_0, F_i \rangle$ be

$$\begin{aligned} S &= \{s_0, s_1\} , \\ \Sigma &= \{a\} , \\ \Delta_i(s_0, a) &= \begin{cases} (s_1, s_0) & i = 3 \\ (s_0, s_1) & i \neq 3, \end{cases} \\ \Delta_i(s_1, a) &= (s_1, s_1) \text{ for } i = 1, \dots, 6 , \\ F_i &= \begin{cases} \{s_1\} & i = 3 \\ \{s_0\} & i \neq 3 . \end{cases} \end{aligned}$$

Then $L_i(M_i) = \emptyset$ for each $i=1, \dots, 6$. However, $L_i(M_i) \neq \emptyset$ for each $i=1, \dots, 6$.

REMARK. In the proof of Theorem 3.4.1, $|\tilde{\mathcal{F}}| = |\mathcal{F}|$ for $i=1, \dots, 5$. However, for the C_6 -acceptance, the size of $\tilde{\mathcal{F}}$ is increased in the construction of $\tilde{M}$.

The following embedding theorem establishes the relationship between $\mathcal{N}_i$ (resp. $\mathcal{D}_i$) and $\mathcal{NT}_i$ (resp. $\mathcal{DT}_i$) for $i=1, \dots, 6$.

THEOREM 3.4.2. For $i=1, \dots, 6$ and $L \subseteq \Sigma^{T_1}$, L is in $\mathcal{N}_i$ (resp. $\mathcal{D}_i$) if and only if $\iota(L)$ is $\mathcal{NT}_i$ (resp. $\mathcal{DT}_i$).

PROOF. For $L \subseteq \Sigma^{T_1}$, let $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ be a finite automaton such that $L = L_i(M)$. Then a finite tree automaton $\tilde{M} = \langle \tilde{S}, \Sigma, \tilde{\Delta}, \tilde{s}_0, \tilde{\mathcal{F}} \rangle$ which accepts $\iota(L)$ in the sense of C_i is constructed as follows:

- (1) $\tilde{S} = S \cup \{\perp, \top\}$, where $\perp, \top \notin S$.
- (2) $\tilde{s}_0 = s_0$.
- (3) For each s in S ,

$$\begin{aligned} \tilde{\Delta}(s, \sigma) &= \{(p, \top) \mid p \in \Delta(s, \sigma)\} \text{ for } \sigma \text{ in } \Sigma , \\ \tilde{\Delta}(\top, a) &= \{(\top, \top)\} , \quad \Delta(\top, b) = \{(\perp, \perp)\} , \\ \tilde{\Delta}(\perp, \sigma) &= \{(\perp, \perp)\} \text{ for } \sigma \text{ in } \Sigma . \end{aligned}$$

$$(4) \quad \tilde{\mathcal{F}} = \begin{cases} \mathcal{F} \cup \{\{\top\}\} & \text{for } i = 1, 2, 3, 5 \\ \mathcal{F} \cup \{\{s_0, \top\}\} & \text{for } i = 4, 6 \end{cases}$$

Notice that if M is deterministic then so is $\tilde{M}$.

The converse direction follows from Theorem 3.4.1.

3.5. Comparison of the classes concerned

This section relates the classes $\mathcal{NF}_i$ and $\mathcal{DF}_i$ for $i=1, \dots, 6$. The results are summarized in the lattice diagram in Fig. 3-1, where an unbroken line indicates a proper containment and a broken line means a containment whose properness is unknown. One can also draw a similar diagram for the deterministic classes.

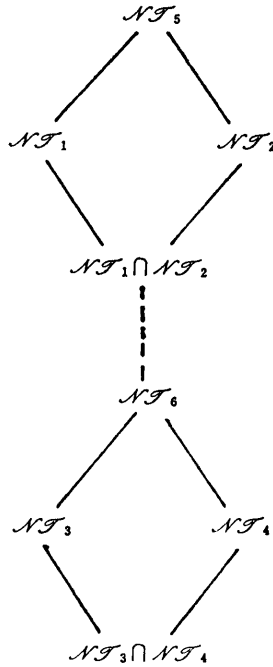


Fig. 3-1

NOTATION. Let t be in Σ^{T_2} . For a path $\pi = \{x_0 < x_1 < \dots x_n < \dots\}$ in T_2 , we denote by $[t|\pi] \in \Sigma^{T_1}$ a function defined by $[t|\pi](0^n) = t(x_n)$ for $n \geq 0$.

PROPOSITION 3.5.1. (1) $\mathcal{NF}_1$ and $\mathcal{NF}_2$ are incomparable.

(2) $\mathcal{DT}_1$ and $\mathcal{DT}_2$ are incomparable.

PROOF. Let $\Sigma = \{a, b\}$ and let

$$L_1 = \iota(L'), \text{ where } L' = (ba^*)^a \subseteq \Sigma^{T_1},$$

$$L_2 = \{t \in \Sigma^{T_2} \mid [t] \pi \in \Sigma^* a^a \subseteq \Sigma^{T_1} \text{ for every path } \pi \text{ in } T_2\}.$$

We can easily construct a deterministic finite tree automaton which accepts L_i in the sense of C_i for $i=1, 2$. Since L' is known not to be in F_σ^R [20, 33] and since $\mathcal{NT}_2 = F_\sigma^R$ from Table 1-1 of Chapter 1, it follows from Theorem 3.4.2 that L_1 is not in $\mathcal{NT}_2$. On the other hand, Rabin [28] has proved that L_2 is not in $\mathcal{NT}_1$. Hence L_1 is in $\mathcal{DT}_1 - \mathcal{NT}_2$ and L_2 is in $\mathcal{DT}_2 - \mathcal{NT}_1$.

PROPOSITION 3.5.2. (1) $\mathcal{NT}_3$ and $\mathcal{NT}_4$ are incomparable.

(2) $\mathcal{DT}_3$ and $\mathcal{DT}_4$ are incomparable.

PROOF. These results follow from Theorem 3.4.2 and the fact that G^R and F^R in Table 1-1 of Chapter 1 are incomparable.

THEOREM 3.5.3. (1) $\mathcal{NT}_3 \cup \mathcal{NT}_4 \subseteq \mathcal{NT}_6$.

(2) $\mathcal{DT}_3 \cup \mathcal{DT}_4 \subseteq \mathcal{DT}_6$.

PROOF. By Theorem 3.3.1, we deal with a simple finite tree automaton $M = \langle S, \Sigma, \Delta, s_0, F \rangle$. Then we consider finite tree automata $M_0 = \langle S, \Sigma, \Delta, s_0, \mathcal{F}_0 \rangle$ and $M_1 = \langle S, \Sigma, \Delta, s_0, \mathcal{F}_1 \rangle$, where $\mathcal{F}_0 = \{G \mid G \cap F \neq \emptyset, G \subseteq S\}$ and $\mathcal{F}_1 = \{G \mid \emptyset \neq G \subseteq F\}$. Then it is not difficult to see that $L_3(M) = L_6(M_0)$ and $L_4(M) = L_6(M_1)$. By the definition, it is obvious that if M is deterministic then so are M_0 and M_1 . The properness of the inclusions follows from Theorem 3.4.2 and the fact that $G^R \cup F^R \subseteq G_\sigma^R \cap F_\sigma^R$ in Table 1-1 of Chapter 1.

THEOREM 3.5.4. [23]. (1) $\mathcal{NT}_6 \subseteq \mathcal{NT}_1 \cap \mathcal{NT}_2$.

(2) $\mathcal{DT}_6 \subseteq \mathcal{DT}_1 \cap \mathcal{DT}_2$.

PROOF. Let $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ be a finite tree automaton. We construct a finite tree automaton $M' = \langle S', \Sigma, \Delta', s'_0, F' \rangle$ such that $L_6(M) = L_1(M') = L_2(M')$ as follows:

(1) $S' = S \times P(S)$.

(2) $s'_0 = (s_0, \{s_0\})$.

- (3) For each $s \in S$, $Q \subseteq S$ and $\sigma \in \Sigma$, $\mathcal{A}'((s, Q), \sigma) = \{((p, Q_1), (q, Q_2)) \mid (p, q) \in \mathcal{A}(s, \sigma), Q_1 = Q \cup \{p\}, Q_2 = Q \cup \{q\}\}$.
- (4) $F' = \{(s, F) \mid s \in S, F \in \mathcal{F}\}$.

As simulating M , M' stores the states occurring on the path from the root to the current node in the second component of its state. Let t be in $L_1(M')$ and let r' be a C_1 -accepting run of M' on t . Then we define r by $r(x) = p_1(r'(x))$, where p_1 is the projection to the first component. By (3), r is a run of M on t . We show that r is a C_6 -accepting run of M on t . For an arbitrary path $\pi = \{x_0 < x_1 < \dots x_n < \dots\}$ in T_2 , let $r'(x_n) = (s_n, F_n)$ for $n \geq 0$. Then by definition, we see that $F_n \subseteq F_{n+1}$ for $n \geq 0$. Therefore there exists an integer n_0 such that $F_{n_0} = F_n$ for all $n \geq n_0$. Note that $F_{n_0} = \{s_n \mid n \geq 0\}$. Since r' is a C_1 -accepting run, $I(r' \mid \pi) \cap F' \neq \emptyset$. Therefore there is (s, F) in F' such that $(s, F) = (s_n, F_n)$ for infinitely many $n \geq 0$. Since $F_{n_0} = F_n$ for all $n \geq n_0$, $F = F_{n_0}$. Since $F_{n_0} = r(\pi)$, $r(\pi)$ is in $\mathcal{F}$. Hence r is a C_6 -accepting run of M on t . Thus $L_1(M') \subseteq L_6(M)$.

Conversely, let t be in $L_6(M)$ and let r be a C_6 -accepting run of M on t . We define r' inductively. Let $r'(\varepsilon) = (s_0, \{s_0\})$. If $r'(x) = (s, Q)$ is defined, then let $r'(x0) = (r(x0), Q \cup \{r(x0)\})$ and $r'(x1) = (r(x1), Q \cup \{r(x1)\})$. Then r' is a run of M' on t since r is a run of M on t . Let $\pi = \{x_0 < x_1 < \dots x_n < \dots\}$ be an arbitrary path in T_2 . Let $r'(x_n) = (s_n, F_n)$ for $n \geq 0$. Since r is a C_6 -accepting run, $r(\pi) = F$ for some F in $\mathcal{F}$. Since $F_n \subseteq F_{n+1}$ and $F_n = \{s_i \mid 0 \leq i \leq n\}$, there exists an integer n_0 such that $F = F_n$ for all $n \geq n_0$. Since $\{s_n \mid n \geq n_0\}$ is finite, there exists s in S such that $(s, F) = (s_n, F_n)$ for infinitely many $n \geq 0$. Therefore $I(r' \mid \pi) \cap F' \neq \emptyset$. Thus r' is a C_1 -accepting run of M' on t . Hence $L_6(M) \subseteq L_1(M')$. Thus $L_6(M) = L_1(M')$.

We now prove $L_2(M') = L_6(M)$. We use the same definitions as those in the proof of $L_1(M') = L_6(M)$. Suppose that r' is a C_2 -accepting run of M' on t . Then for every path π , $I(r' \mid \pi) \subseteq F'$. Since $I(r' \mid \pi) \neq \emptyset$, there exists (s, F) in F' such that $(s, F) = (s_n, F_n)$ for infinitely many $n \geq 0$. Since $F_n \subseteq F_{n+1}$ for $n \geq 0$, there exists an integer n_0 such that $F = F_n$ for all $n \geq n_0$. Therefore the run r defined by $r(x) = p_1(r'(x))$ for x in T_2 has the property that $r(\pi) = F$. Thus r is a C_6 -accepting run of M on t . Hence $L_2(M') \subseteq L_6(M)$. To see the converse inclusion, it may be sufficient to note that in the proof of $L_6(M) \subseteq L_1(M')$ we have $(s_n, F_n) = (s_n, F)$ for all $n \geq n_0$, where F is in $\mathcal{F}$. Thus $I(r' \mid \pi) \subseteq F'$. Therefore r' is also a C_2 -accepting run of M' on t . Hence $L_6(M) \subseteq L_2(M')$.

By the construction, it is also obvious that if M is deterministic then so is M' .

REMARK. It is an open problem whether the inclusion is proper in Theorem 3.5.4.

THEOREM 3.5.5. (1) $\mathcal{NT}_1 \cup \mathcal{NT}_2 \subsetneq \mathcal{NT}_5$.

(2) $\mathcal{DT}_1 \cup \mathcal{DT}_2 \subsetneq \mathcal{DT}_5$.

PROOF. Given a finite tree automaton $M = \langle S, \Sigma, \Delta, s_0, F \rangle$, we define in the same way as Theorem 3.5.3 finite tree automata $M_0 = \langle S, \Sigma, \Delta, s_0, \mathcal{F}_0 \rangle$ and $M_1 = \langle S, \Sigma, \Delta, s_0, \mathcal{F}_1 \rangle$, where $\mathcal{F}_0 = \{G \mid G \cap F \neq \emptyset, G \subseteq S\}$ and $\mathcal{F}_1 = \{G \mid \emptyset \neq G \subseteq F\}$. Then we can see that $L_1(M) = L_5(M_0)$ and $L_2(M) = L_5(M_1)$.

We now prove the properness of the inclusions. Let $\Sigma = \{a, b\}$ and let

$$L_1 = \iota(L'), \text{ where } L' = (ba^*)^* \subseteq \Sigma^{T_1},$$

$$L_2 = \{t \in \Sigma^{T_2} \mid [t|_{\pi}] \in a\Sigma^*a^o \text{ for every path } \pi \subseteq T_2\}.$$

We will show that $L = L_1 \cup L_2$ is not in $\mathcal{NT}_1 \cup \mathcal{NT}_2$. Assume that L is in $\mathcal{NT}_1$. Let $M = \langle S, \Sigma, \Delta, s_0, F \rangle$ be a finite tree automaton accepting L in the sense of C_1 . One can modify M to get a finite tree automaton M_2 which accepts L_2 in the sense of C_1 since for each $t \in L_1$, $t(\varepsilon) = b$ and on the other hand for each $t \in L_2$, $t(\varepsilon) = a$. By the result due to Rabin [28], L_2 is not in $\mathcal{NT}_1$. Hence L is not in $\mathcal{NT}_1$. Now assume that L is in $\mathcal{NT}_2$. Then by the same consideration, one can get a finite tree automaton which accepts L_1 in the sense of C_2 . In the same way as Proposition 3.5.1, this yields a contradiction. Hence L is not in $\mathcal{NT}_2$. Therefore L is not in $\mathcal{NT}_1 \cup \mathcal{NT}_2$.

We can easily construct a deterministic finite tree automaton which accepts L in the sense of C_5 .

3.6. Structural measures for finite tree automata

In this section we introduce two structural measures for finite tree automata to give further observations on $\mathcal{NT}_i$ for $i=1, \dots, 6$. Let $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ be a finite tree automaton. The first measure is the *nondeterministic degree* of M , denoted by $\text{ndeg}(M)$, which is defined to be $\max\{|\Delta(s, \sigma)| \mid s \in S, \sigma \in \Sigma\}$. The second measure is the number $|\mathcal{F}|$ of final sets. As shown in Section 3.3, the size of $\mathcal{F}$ does not affect the C_i -acceptances for $i=1, \dots, 4$.

DEFINITION 3.6.1. For $i=1, \dots, 6$ and an integer $k \geq 1$, a set L in $\mathcal{NF}_i$ is said to be of *nondeterministic degree* k if $k = \min\{ndeg(M) \mid L = L_i(M)\}$. Then we define

$$k\text{-}\mathcal{NF}_i = \{L \in \mathcal{NF}_i \mid L \text{ is of nondeterministic degree } l \text{ with } 1 \leq l \leq k\}.$$

A deterministic finite automaton is of nondeterministic degree one. The following theorem refines the result in [30] which separates the nondeterministic classes from the deterministic classes.

THEOREM 3.6.1. For $i=1, \dots, 6$ and for $k \geq 1$, $k\text{-}\mathcal{NF}_i \subseteq (k+1)\text{-}\mathcal{NF}_i$.

PROOF. Let $\Sigma = \{a, b\}$. For $n \geq 1$, let t_n be the Σ -tree defined by

$$t_n(x) = \begin{cases} b & \text{if } |x| = n \\ a & \text{otherwise.} \end{cases}$$

Let $L_k = \{t_1, \dots, t_k\}$. We show that L_{k+1} is in $(k+1)\text{-}\mathcal{NF}_i$ but not in $k\text{-}\mathcal{NF}_i$ for $i=1, \dots, 6$. One can easily construct a nondeterministic finite tree automaton of nondeterministic degree $k+1$ which accepts L_{k+1} in the sense of C_i . Suppose that L_{k+1} is accepted in the sense of C_i by a nondeterministic finite tree automaton $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ with $ndeg(M) \leq k$. Let r_n be a C_i -accepting run of M on t_n for $n=1, \dots, k+1$. Since $|\Delta(s_0, a)| \leq k$, there exist m, n , $1 \leq m < n \leq k+1$, such that $r_m(\varepsilon) = r_n(\varepsilon) = s_0$ and $(r_m(0), r_m(1)) = (r_n(0), r_n(1))$. Let t be a Σ -tree defined by

$$t(x) = \begin{cases} b & \text{if either } |x| = m \text{ and } 0 \leq x \text{ or } |x| = n \text{ and } 1 \leq x \\ a & \text{otherwise.} \end{cases}$$

Then from r_m and r_n we can construct a C_i -accepting run of M on t in the following way. We prune the subtree at the node 1 from r_m and instead of it we graft the subtree of r_n at the node 1. Then the resulting run is also a C_i -accepting run of M on t . However, t does not belong to L_{k+1} , a contradiction. Hence L_{k+1} is not in $k\text{-}\mathcal{NF}_i$.

DEFINITION 3.6.2. The *Muller degree* of a language L in $\mathcal{NF}_5$, denoted by $mdeg(L)$, is the minimum number of final sets sufficient for its acceptance in the sense of C_5 , i.e.

$$mdeg(L) = \min\{k \mid L = L_5(M) \text{ for } M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle \text{ with } k = |\mathcal{F}|\}.$$

THEOREM 3.6.2. For each $k \geq 0$, there exists a language L_k in $\mathcal{NF}_5$ such that $k = mdeg(L_k)$.

PROOF. For $n \geq 1$, let t_n be the Σ -tree such that $[t_n | \pi] = (a^n b)^\omega$ for each path π in T_2 . Let $L_0 = \phi$ and $L_k = \{t_1, \dots, t_k\}$ for $k \geq 1$. It is clear that $mdeg(L_0) = 0$. We will prove that $mdeg(L_k) = k$ for $k \geq 1$. One can easily construct a deterministic finite tree automaton $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ such that $k = |\mathcal{F}|$ and $L_k = L_s(M)$. Thus $mdeg(L_k) \leq k$. Assume that $mdeg(L_k) = m < k$. Then there exists a nondeterministic finite tree automaton $\tilde{M} = \langle \tilde{S}, \Sigma, \tilde{\Delta}, \tilde{s}_0, \tilde{\mathcal{F}} \rangle$ such that $m = |\tilde{\mathcal{F}}|$ and $\rho(L_k) = L_s(\tilde{M})$ because the construction of $\tilde{M}$ from M preserves the number of final sets. Since $\rho(L_k) = (ab)^\omega \cup (a^2b)^\omega \cup \dots \cup (a^k b)^\omega$ and $m < k$, there exist $i \neq j$ such that both $(a^i b)^\omega$ and $(a^j b)^\omega$ can be accepted with the same final set F in $\tilde{\mathcal{F}}$. Then there is $w \in \Sigma^*$ such that $a^i b w (a^j b)^\omega$ can be accepted by $\tilde{M}$ with F in the sense of C_5 . This is a contradiction. Hence $mdeg(L_k) = k$.

REMARK. Theorem 3.6.2 still holds when we restrict the case to the deterministic C_5 -acceptance. By the results in Section 3.3, we can also define a similar degree for a language in $\mathcal{NT}_6$. Although the language L_k defined in the proof of Theorem 3.6.2 also seems to be a candidate which has exactly degree k in this case, we have not proved this yet.

3.7. Concluding remarks

By the results in Sections 3.4-3.6, the class $\mathcal{NT}_5$ that is studied in [27] is the largest one among the classes defined by finite tree automata. This class is closed under Boolean operations [27] and for each L in $\mathcal{NT}_5$, $\rho(L)$ is an ω -regular language. Hence the class $\mathcal{NT}_5$ may be considered to be a natural extension of the class of ω -regular languages to infinite tree languages.

In [23], Moriya has characterized $\mathcal{NT}_3$ and $\mathcal{NT}_4$ in terms of general topology on Σ^{T^2} . We have not considered this approach in detail.

We have ignored the classes $\mathcal{N}_i^{(k)}$ for $k > 2$. However, the arguments in this paper are easily applicable to $\mathcal{N}_i^{(k)}$ and we can get similar results for $\mathcal{N}_i^{(k)}$ for $k > 2$. For instance, Theorems 3.4.1. and 3.4.2 can be generalized as follows: Let $\iota_k: \Sigma^{T_k} \rightarrow \Sigma^{T_{k+1}}$ be a function defined by $\iota_k(t)(x) = t(x)$ if x is in T_k and $\iota_k(t)(x) = a$ if x is in $T_{k+1} - T_k$. Let $\rho_k: \Sigma^{T_{k+1}} \rightarrow \Sigma^{T_k}$ be a function defined by $\rho_k(t) = t|_{T_k}$ for t in $\Sigma^{T_{k+1}}$. Then we have the following generalizations:

THEOREM 3.4.1'. *If L is in $\mathcal{A}_i^{(k+1)}$, then $\rho_k(L)$ is in $\mathcal{A}_i^{(k)}$ for $i=1, \dots, 6$.*

THEOREM 3.4.2'. *L is in $\mathcal{A}_i^{(k)}$ if and only if $\iota_k(L)$ is in $\mathcal{A}_i^{(k+1)}$ for $i=1, \dots, 6$.*

Chapter 4. Finite biautomata on two-sided infinite words

4.1. Introduction

We define a two-sided infinite word as the equivalence class under the shift of a biinfinite word (i.e. a mapping from the set of integers $\mathbb{Z}$ into a finite alphabet Σ). Finite biautomata on this kind of words were firstly investigated by Nivat and Perrin [25]. They represent a biinfinite word as an element in $\Sigma^\omega \times \Sigma^\omega$. A finite biautomaton is defined to be a triple consisting of two finite automata and a subset made of pairs whose first (resp. second) component is a state of the first (resp. second) finite automata. This set is called an initial set of a finite biautomaton. Given a biinfinite word (x, y) in $\Sigma^\omega \times \Sigma^\omega$, one of pairs in the initial set is chosen and the first (resp. second) finite automaton runs on x (resp. y) infinitely starting with the first (resp. second) component of the pair as its initial state.

Nivat and Perrin defined both nondeterministic and deterministic finite biautomata through one kind of acceptance condition due to Büchi [2] and called them bilateral if the biinfinite languages accepted are closed under the shift operation. Thus a bilateral finite biautomaton becomes an acceptor for two-sided infinite languages. In this chapter we study both nondeterministic and deterministic finite biautomata under six types of acceptance conditions discussed in Chapters 1-3. A finite biautomaton is called strict if its initial set is a singleton. We show that strict bilateral finite biautomata are less powerful than bilateral finite biautomata under each acceptance condition. We investigate interrelationship of classes of two-sided infinite languages defined by bilateral finite biautomata. Characterization of several classes of two-sided infinite languages is also given.

4.2. Preliminaries

DEFINITION 4.2.1. Let Σ be a finite alphabet. Let $\Sigma^\infty = \Sigma^* \cup \Sigma^\omega$. We call (x, y) in $\Sigma^\infty \times \Sigma^\infty$ a *biword*. A biword (x, y) is said to be: finite if

$(x, y) \in \Sigma^* \times \Sigma^*$; right-infinite if $(x, y) \in \Sigma^* \times \Sigma^\omega$; left-infinite if $(x, y) \in \Sigma^\omega \times \Sigma^*$; *biinfinite* if $(x, y) \in \Sigma^\omega \times \Sigma^\omega$. We define an equivalence relation denoted by $\sim$ over $\Sigma^\omega \times \Sigma^\omega$ as:

For (x, y) and (x', y') in $\Sigma^\omega \times \Sigma^\omega$, $(x, y) \sim (x', y')$ if and only if there exists u in Σ^* such that either $y = uy'$ and $x' = u^\rho x$ or $y' = uy$ and $x = u^\rho x'$, where ρ is the reversal operator of Σ^* .

An equivalence class of biwords under $\sim$ is called a *two-sided word*. The set of two-sided words is denoted by ${}^\omega\Sigma^\omega$, and $\pi: \Sigma^\omega \times \Sigma^\omega \rightarrow {}^\omega\Sigma^\omega$ is the canonical surjection. The set of two-sided finite (resp. right-infinite) words can be identified with Σ^* (resp. Σ^ω). We denote the set of two-sided left-infinite words by ${}^\omega\Sigma$. One can define a bijection between ${}^\omega\Sigma$ and Σ^ω by

$$z = \pi(x, y) \rightarrow z^\rho = y^\rho x.$$

Moreover, one can define the product of a word x in ${}^\omega\Sigma$ with a word y in Σ^ω as the equivalence class of the pair (x^ρ, y) , which is denoted by xy .

The set of two-sided biinfinite words ${}^\omega\Sigma^\omega = \Sigma^\omega \times \Sigma^\omega / \sim$ is main concern of our study. We simply call x in ${}^\omega\Sigma^\omega$ a *two-sided infinite word*. A subset $L \subseteq {}^\omega\Sigma^\omega$ is called a *two-sided infinite language*.

For acceptors of two-sided infinite languages, Nivat and Perrin [25] introduce the notion of finite biautomata.

DEFINITION 4.2.2. A *nondeterministic finite biautomaton* (abbreviated *nba*) is a triple $M = \langle M_-, M_+, H \rangle$, where

- (1) M_- and M_+ are *nfa*'s with $M_- = \langle S_-, \Sigma, \Delta_-, s_{0-}, \mathcal{F}_- \rangle$ and $M_+ = \langle S_+, \Sigma, \Delta_+, s_{0+}, \mathcal{F}_+ \rangle$.
- (2) H is a subset of $S_- \times S_+$ called the *initial set*. If $H = \{(s_{0-}, s_{0+})\}$, then we say that M is *strict*.

An *nba* $M = \langle M_-, M_+, H \rangle$ is said to be *deterministic* (abbreviated *dba*) if both M_- and M_+ are deterministic finite automata.

Intuitively, for (x, y) in $\Sigma^\omega \times \Sigma^\omega$, M_- runs on x infinitely starting with a state s_- and M_+ runs on y infinitely starting with a state s_+ , where (s_-, s_+) is in H . Thus the initial state s_{0-} (resp. s_{0+}) of M_- (resp. M_+) is in a sense surplus. To remedy this discrepancy, for an *nfa* $M = \langle S, \Sigma, \Delta, s_0, \mathcal{F} \rangle$ and a state s in S , we use notation $\langle M; s \rangle$ to denote the *nfa* obtained from M by regarding s as the initial state. Namely

$$\langle M; s \rangle = \langle S, \Sigma, A, s, \mathcal{F} \rangle.$$

We make use of six types of acceptance conditions (C_i for $i=1, \dots, 6$) introduced in Chapter 1 to define the corresponding acceptance notions for finite biautomata. Therefore the reader is suggested to review Definitions 1.2.3-1.2.5.

DEFINITION 4.2.3. For an *nba* $M = \langle M_-, M_+, H \rangle$, the set of biinfinite words accepted by M in the sense of C_i ($i=1, \dots, 6$), denoted $L_i(M)$, is defined as:

$$L_i(M) = \{ (x, y) \mid (x, y) \in \Sigma^\omega \times \Sigma^\omega, \text{ and there exists } (s_-, s_+) \text{ in } H \text{ such that } x \text{ is in } L_i(\langle M_-; s_- \rangle) \text{ and } y \text{ is in } L_i(\langle M_+; s_+ \rangle) \}.$$

An *nba* M is said to be *bilateral* in the sense of C_i ($i=1, \dots, 6$) if $L_i(M)$ is saturated with the relation $\sim$. If M is bilateral in the sense of C_i , the set of two-sided infinite words recognized by M (denoted $B_i(M)$) is defined as:

$$B_i(M) = \pi(L_i(M)) \subseteq \Sigma^\omega.$$

For a two-sided infinite language $L \subseteq \Sigma^\omega$, L is said to be C_i -recognizable ($i=1, \dots, 6$) if there exists a bilateral *nba* M such that $L = B_i(M)$.

DEFINITION 4.2.4. For $i=1, \dots, 6$, we define

- (1) $\mathcal{NB}_i = \{B_i(M) \mid M \text{ is a bilateral nba}\},$
- (2) $\mathcal{SB}_i = \{B_i(M) \mid M \text{ is a strict bilateral nba}\},$
- (3) $\mathcal{DB}_i = \{B_i(M) \mid M \text{ is a bilateral dba}\},$
- (4) $\mathcal{SDB}_i = \{B_i(M) \mid M \text{ is a strict bilateral dba}\}.$

4.3. Simple finite biautomata

In this section we consider finite biautomata whose two components are simple finite automata. We call this type of finite biautomata *simple*.

THEOREM 4.3.1. Let $M = \langle M_-, M_+, H \rangle$ be an *nba*. Then for each $i=1, \dots, 4$, there exists an *nba* $\tilde{M} = \langle \tilde{M}_-, \tilde{M}_+, \tilde{H} \rangle$ such that $L_i(M) = L_i(\tilde{M})$ and both $\tilde{M}_-$ and $\tilde{M}_+$ are simple. Furthermore, if M is deterministic or strict, so is $\tilde{M}$.

PROOF. Let $M_- = \langle S_-, \Sigma, A_-, s_{0-}, \mathcal{F}_- \rangle$ and $M_+ = \langle S_+, \Sigma, A_+, s_{0+}, \mathcal{F}_+ \rangle$.

Case 1. $i=1, 3$: $\tilde{M} = \langle \tilde{M}_-, \tilde{M}_+, \tilde{H} \rangle$ is simply defined by $\tilde{M}_- = \langle S_-, \Sigma, A_-, s_{0-}, \tilde{F}_- \rangle$, $\tilde{M}_+ = \langle S_+, \Sigma, A_+, s_{0+}, \tilde{F}_+ \rangle$ and $\tilde{H} = H$, where $\tilde{F}_- = \{s \mid s \in F \text{ for}$

some F in $\mathcal{F}_-$ and $\tilde{F}_+ = \{s | s \in F \text{ for some } F \text{ in } \mathcal{F}_+\}$.

Case 2. $i=2$: $\tilde{M} = \langle \tilde{M}_-, \tilde{M}_+, H \rangle$ is defined as follows: Make a simple finite automaton $\tilde{M}_-$ (resp. $\tilde{M}_+$) from M_- (resp. M_+) in the same way as done in the proof of Theorem 3.3.1 (Case 2). Let

$$\tilde{H} = \{((s_-, \phi), (s_+, \phi)) | (s_-, s_+) \in H\}.$$

By the similar argument done in the proof of Theorem 3.3.1 (Case 2), we can easily show that $L_2(M) = L_2(\tilde{M})$.

Case 3. $i=4$: $\tilde{M} = \langle \tilde{M}_-, \tilde{M}_+, \tilde{H} \rangle$ is defined as follows: Make a simple finite automaton $\tilde{M}_-$ (resp. $\tilde{M}_+$) from M_- (resp. M_+) in the same way as done in the proof of Theorem 3.3.1 (Case 3). Let

$$\begin{aligned} \tilde{H} = \{ & ((s_-, \mathcal{F}_0), (s_+, \mathcal{F}_1)) | (s_-, s_+) \in H, \mathcal{F}_0 = \{F | s_- \in F, F \in \mathcal{F}_-\} \text{ and} \\ & \mathcal{F}_1 = \{F | s_+ \in F, F \in \mathcal{F}_+\} \} . \end{aligned}$$

By the similar argument done in the proof of Theorem 3.3.1 (Case3), we can easily show that $L_4(M) = L_4(\tilde{M})$.

In Cases 1-3, the construction of $\tilde{M}$ shows that if M is deterministic or strict then so is $\tilde{M}$.

4.4. Strict bilateral finite biautomata are less powerful

In this section we show that strict bilateral finite biautomata are less powerful than bilateral finite biautomata under each acceptance condition C_i for $i=1, \dots, 6$. Note that by Theorem 4.3.1, it is sufficient to consider only simple finite biautomata for $i=1, \dots, 4$.

THEOREM 4.4.1. *For each $i=1, \dots, 6$,*

- (1) $\mathcal{SAB}_i \subseteq \mathcal{NAB}_i$,
- (2) $\mathcal{SDB}_i \subseteq \mathcal{DB}_i$.

PROOF. We fix $\Sigma = \{a, b\}$ in this proof. From Definition 4.2.4, the containment is evident in each case. We will prove the properness in the following:

Case 1. $i=1, 2$: Consider the two-sided infinite language $L_1 = {}^*ab^*$. We can make a bilateral *dba* $M = \langle M_-, M_+, H \rangle$ such that $L_1 = B_1(M)$ as illustrated in Fig. 4-1:

Suppose that there exists a strict bilateral *nba* $M' = \langle M'_-, M'_+, \{(s'_{-}, s'_{+})\} \rangle$ such that $L_1 = B_1(M')$.

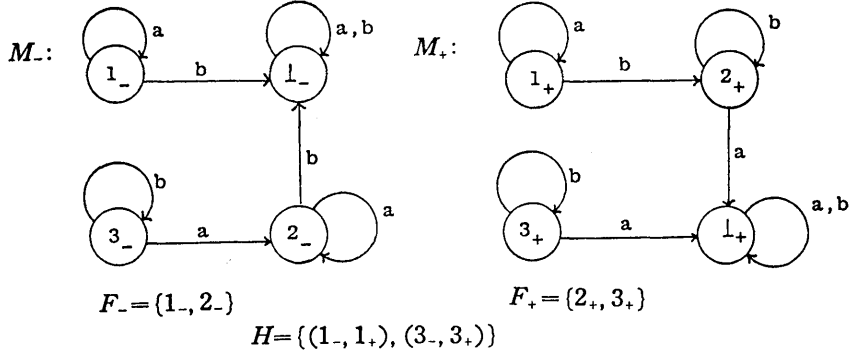


Fig. 4-1

(1) For ab^ω , if we let M' start at a 's parts, i.e. M'_- runs on a^ω and M'_+ runs on a^nb^ω for some $n > 0$ with initial states s'_{0-} and s'_{0+} respectively, then there exist a C_1 -accepting run of M'_- on a^ω and that of M'_+ on a^nb^ω .

(2) In the similar way, if we let M' start at b 's parts, i.e. M'_- runs on b^na^ω for some $n > 0$ and M'_+ runs on b^ω with initial states s'_{0-} and s'_{0+} respectively, then there exist a C_1 -accepting run of M'_- on b^na^ω and that of M'_+ on b^ω .

From (1) and (2), we can get a C_1 -accepting run of M'_+ on a^nb^ω and that of M'_- on b^na^ω . This means that $(b^na^\omega, a^nb^\omega)$ is accepted by M' . Since M' is bilateral, $ab^na^nb^\omega$ is in $B_1(M')$. A contradiction. Since $\mathcal{SBD}_1 \subseteq \mathcal{SBDN}_1$, we have shown properness of both deterministic and non-deterministic cases.

Note that this argument is also applicable for C_2 -acceptance and $L_1 = L_2(M)$. Thus we have also done for $i=2$.

Case 2. $i=3$: Let $L_3 = \Sigma^*b\Sigma^*$. We can easily construct a bilateral dba M such that $L_3 = L_3(M)$. Assume that there exists a strict bilateral nba M' such that $L_3 = L_3(M')$. Consider aba^ω .

(3) If we let M' start at left a 's parts, i.e. M'_- runs on a^ω and M'_+ runs on a^mba^ω for some $m > 0$ with initial states s'_{0-} and s'_{0+} respectively, then there exist a C_3 -accepting run of M'_- on a^ω and that of M'_+ on a^mba^ω .

(4) In the similar way, if we let M' start at right a 's parts, i.e. M'_- runs on a^mba^ω for some $m > 0$ and M'_+ runs on a^ω with initial states s'_{0-} and s'_{0+} respectively, then there exist a C_3 -accepting run of M'_- on

a^nba^w and that of M'_+ on a^w .

From (3) and (4), we can get a C_3 -accepting run of M'_+ on a^w and that of M'_- on a^w . This means that (a^w, a^w) is accepted by M' . Since M' is bilateral, a^w is in L_3 . A contradiction.

Case 3. $i=4$: Let $L_4=\{^wa^w, ^wb^w\}$. A bilateral dba which accepts L_4 in the sense of C_4 can be easily constructed. Suppose that there exists a strict bilateral nba M such that $L_4=B_4(M)$. Then from a C_4 -accepting run of M for $^wa^w$ and from that for $^wb^w$, we can get a C_4 -accepting run of M for $^wab^w$ (or $^wba^w$). This is a contradiction.

Case 4. $i=5$: Let $L_5=^wb(a^*b)^w \cup ^wa\sum^*b^w$. We can make a bilateral dba $M=\langle M_-, M_+, H \rangle$ such that $L_5=B_5(M)$ as illustrated below:

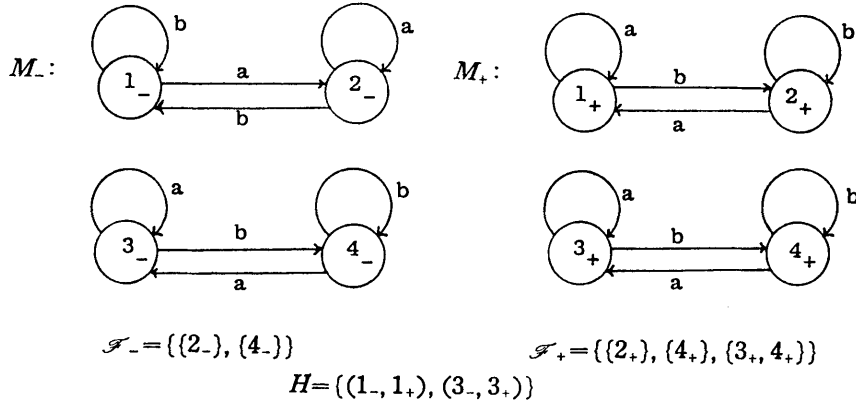


Fig. 4-2

Suppose that there exists a strict bilateral nba $M'=\langle M'_-, M'_+, \{(s'_{0-}, s'_{0+})\} \rangle$ such that $L_5=B_5(M')$.

(5) For $^wb(ab)^w \in L_5$, we let M' start at b 's parts, i.e. M'_- runs on b^w and M'_+ runs on $b^n(ab)^w$ for some $n \geq 0$.

(6) For $^wab^w \in L_5$, we let M' start at a 's parts, i.e. M'_- runs on a^w and M'_+ runs on $a^n b^w$ for some $n \geq 0$.

From (5) and (6), we can get a C_5 -accepting run of M'_+ on $b^n(ab)^w$ and that of M'_- on a^w . This means that $^wab^n(ab)^w$ is in $L_5=B_5(M')$. A contradiction.

Case 5: $i=6$: Let $L_6=^waba^w$. We can make a bilateral dba $M=\langle M_-, M_+, H \rangle$ such that $L_6=B_6(M)$ as illustrated in Fig. 4-3:

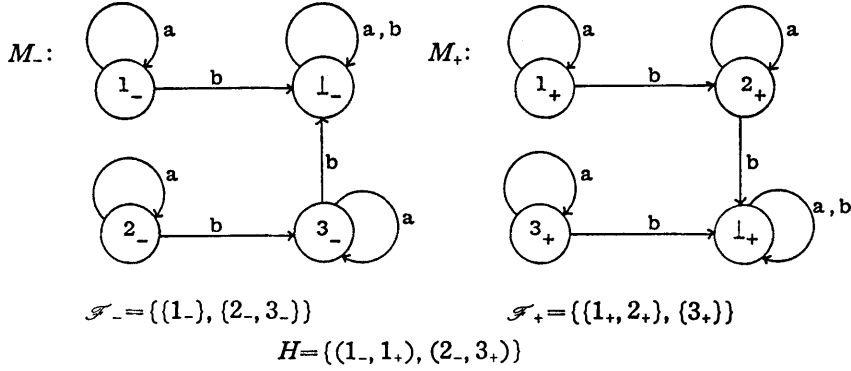


Fig. 4-3

Existence of a strict bilateral *nba* M such that $L_s = L_s(M)$ leads to a contradiction can be shown similarly in Case 2.

REMARK. This theorem is interesting because strictness gives no weakness in definability of ω -languages by finite automata. Namely even when we extend the notion of finite automata to allow multiple initial states, the class of ω -languages defined under C_i (for $i=1, \dots, 6$) acceptance condition is unchanged. This is proved by direct product construction of 'union' machine of two finite automata.

4.5. Equivalence of several classes

In this section we show that some classes of two-sided infinite languages are equivalent. For this purpose we make use of characterization of classes of *biinfinite languages*, i.e. subsets of $\Sigma^\omega \times \Sigma^\omega$, defined by finite biautomata.

DEFINITION 4.5.1. For $i=1, \dots, 6$, we define

- (1) $\mathcal{N}_i(\Sigma^\omega \times \Sigma^\omega) = \{L_i(M) \mid M \text{ is an } nba\}$,
- (2) $\mathcal{SN}_i(\Sigma^\omega \times \Sigma^\omega) = \{L_i(M) \mid M \text{ is a strict } nba\}$,
- (3) $\mathcal{D}_i(\Sigma^\omega \times \Sigma^\omega) = \{L_i(M) \mid M \text{ is a } dba\}$,
- (4) $\mathcal{SD}_i(\Sigma^\omega \times \Sigma^\omega) = \{L_i(M) \mid M \text{ is a strict } dba\}$.

In what follows we use the classes $\mathcal{D}_i$ and $\mathcal{N}_i$ ($i=1, \dots, 6$) of ω -languages and their characterization discussed in Section 1.3.

NOTATION. For two classes of ω -languages $\mathcal{L}$ and $\mathcal{L}'$, $\mathcal{L} \times \mathcal{L}' = \{(L, L')\}$

$\{L \in \mathcal{L} \text{ and } L' \in \mathcal{L}'\}$.

The following proposition is an immediate consequence of Definition 4.5.1.

PROPOSITION 4.5.1. *For $i=1, \dots, 6$, we have*

- (1) $\mathcal{S}\mathcal{N}_i(\Sigma^\omega \times \Sigma^\omega) = \mathcal{N}_i \times \mathcal{N}_i$,
- (2) $\mathcal{S}\mathcal{D}_i(\Sigma^\omega \times \Sigma^\omega) = \mathcal{D}_i \times \mathcal{D}_i$.

From Proposition 4.5.1 and Table 1-1, we get the following characterization table.

Table 4-1

	C_1	C_2	C_3	C_4	C_5	C_6
$\mathcal{S}\mathcal{D}_i(\Sigma^\omega \times \Sigma^\omega)$	$G_\theta^R \times G_\theta^R$	$F_\theta^R \times F_\theta^R$	$G^R \times G^R$	$F^R \times F^R$	$R \times R$	$(G_\theta^R \cap F_\theta^R) \times (G_\theta^R \cap F_\theta^R)$
$\mathcal{S}\mathcal{N}_i(\Sigma^\omega \times \Sigma^\omega)$	$R \times R$	$F_\theta^R \times F_\theta^R$	$G^R \times G^R$	$F^R \times F^R$	$R \times R$	$F_\theta^R \times F_\theta^R$

COROLLARY 4.5.2.

- (1) $\mathcal{S}\mathcal{NB}_1 = \mathcal{S}\mathcal{NB}_5 = \mathcal{S}\mathcal{DB}_5$.
- (2) $\mathcal{S}\mathcal{NB}_2 = \mathcal{S}\mathcal{DB}_2 = \mathcal{S}\mathcal{NB}_6$.
- (3) $\mathcal{S}\mathcal{NB}_3 = \mathcal{S}\mathcal{DB}_3$.
- (4) $\mathcal{S}\mathcal{NB}_4 = \mathcal{S}\mathcal{DB}_4$.

PROPOSITION 4.5.3. *For $i=1, \dots, 6$, we have*

- (1) $\mathcal{N}_i(\Sigma^\omega \times \Sigma^\omega) = \{X_1 \cup X_2 \cup \dots \cup X_n \mid X_1, \dots, X_n \in \mathcal{S}\mathcal{N}_i(\Sigma^\omega \times \Sigma^\omega), n \geq 0\}$,
- (2) $\mathcal{D}_i(\Sigma^\omega \times \Sigma^\omega) = \{X_1 \cup X_2 \cup \dots \cup X_n \mid X_1, \dots, X_n \in \mathcal{S}\mathcal{D}_i(\Sigma^\omega \times \Sigma^\omega), n \geq 0\}$.

PROOF. For an *nba* (resp. *dba*) $M = \langle M_-, M_+, H \rangle$, we make a strict *nba* (resp. *dba*) $M_{(s_-, s_+)} = \langle \langle M_-; s_- \rangle, \langle M_+; s_+ \rangle, \{(s_-, s_+)\} \rangle$ for each (s_-, s_+) in H . Then $L_i(M) = \bigcup_{(s_-, s_+) \in H} L_i(M_{(s_-, s_+)})$ for each $i=1, \dots, 6$. The reverse inclusion can be shown by making an *nba* (resp. *dba*) as disjoint sum of component strict *nba*'s (resp. *dba*'s).

COROLLARY 4.5.4.

- (1) $\mathcal{NB}_1 = \mathcal{NB}_5 = \mathcal{DB}_5$.
- (2) $\mathcal{NB}_2 = \mathcal{DB}_2 = \mathcal{NB}_6$.
- (3) $\mathcal{NB}_3 = \mathcal{DB}_3$.

$$(4) \quad \mathcal{NB}_4 = \mathcal{DB}_4.$$

4.6. Characterization of several classes

In this section we give characterization theorems for several classes of two-sided infinite languages. For the representations of the two-sided infinite languages in these classes, we need several operations defined below which are ‘reverse’ analog of those defined in Section 1.3. The reader is suggested to review Section 1.3.

DEFINITION 4.6.1. For a language $L \subseteq \Sigma^*$, we define:

(1) ${}^oL = \{x \mid x \in {}^o\Sigma \text{ and each initial segment of } x^o \text{ is in } A(L^o)\}$, where $A(U) = \{w \mid w \in \Sigma^* \text{ and } w \text{ is an initial segment of some word in } U\}$ for a set $U \subseteq \Sigma^*$.

(2) ${}^oL = \{x \mid x \in {}^o\Sigma, x^o \in (L^o)^o\}$.

THEOREM 4.6.1 (Nivat and Perrin [25]). *For $L \subseteq {}^o\Sigma^o$, the following conditions are equivalent.*

- (1) L is in $\mathcal{NB}_1$.
- (2) L is a finite union of sets of the form ${}^oXYZ^o$, where X, Y and Z are regular languages.
- (3) There exists L' in $\mathcal{N}_1(\Sigma^o \times \Sigma^o)$ such that $L = \pi(L')$.
- (4) L is in $\mathcal{NB}_5$.
- (5) L is in $\mathcal{DB}_5$.

PROOF. (1) $\Rightarrow$ (2). Let $L = B_1(M)$. Then $L = \pi(L_1(M))$ and by Propositions 4.5.1 and 4.5.3, $L_1(M)$ is a finite union of sets of the form (UV^o, WZ^o) such that U, V, W and Z are regular languages. If we set $X = V^o$ and $Y = U^oW$, then $\pi(UV^o, WZ^o) = {}^oXYZ^o$.

(2) $\Rightarrow$ (3). Since $\mathcal{N}_1(\Sigma^o \times \Sigma^o)$ is closed under union, it suffices to show that (2) implies (3) for a set of the form $L = {}^oXYZ^o$ such that X, Y and Z are regular languages. Let $\langle M_-; s_- \rangle$ be an *nfa* such that $L_1(\langle M_-; s_- \rangle) = (X^o)^o$ and $\langle M_+; s_+ \rangle$ be an *nfa* such that $L_1(\langle M_+; s_+ \rangle) = YZ^o$. If we choose $H = \{(s_-, s_+)\}$, then the *nba* $M = \langle M_-, M_+, H \rangle$ has the property that $L_1(M) = ((X^o)^o, YZ^o)$ and $L = \pi(L_1(M))$.

(3) $\Rightarrow$ (1). Let $M = \langle M_-, M_+, H \rangle$ be a simple *nba* such that $L = \pi(L_1(M))$. We define an *nba* M' as follows:

$S'_- = S'_+ = S_- \cup S_+ \cup \{\perp\}$, where $\perp$ is a new state,

$F'_- = F_-$, $F'_+ = F_+$,

$H' = H \cup \{(s, s) | s \in S_- \cup S_+\}$;

For each s in $S_- \cup S_+$ and σ in Σ ,

$$A'_-(s, \sigma) = \begin{cases} A_-(s, \sigma) & \text{if } s \in S_- \\ \{q | q \in S_+, s \in A_+(q, \sigma)\} \cup \{q | \exists p \in S_- \text{ s.t. } (p, s) \in H \text{ and } q \in A_-(p, \sigma)\} & \text{if } s \in S_+ \text{ and the right-hand side is not empty} \\ \{\perp\} & \text{otherwise ,} \end{cases}$$

$$A'_-(\perp, \sigma) = \{\perp\} .$$

In the similar way:

$$A'_+(s, \sigma) = \begin{cases} A_+(s, \sigma) & \text{if } s \in S_+ \\ \{q | q \in S_-, s \in A_-(q, \sigma)\} \cup \{q | \exists p \in S_+ \text{ s.t. } (s, p) \in H \text{ and } q \in A_+(p, \sigma)\} & \text{if } s \in S_- \text{ and the right-hand side is not empty} \\ \{\perp\} & \text{otherwise ,} \end{cases}$$

$$A'_+(\perp, \sigma) = \{\perp\} .$$

It is easily seen that if $S_- \cap S_+ = \emptyset$, the *nba* is bilateral and $L = B_1(M)$. (This process of making an *nba* bilateral is called *bilateralization*.) Therefore L is in $\mathcal{NB}_1$.

(1) $\Leftrightarrow$ (4) $\Leftrightarrow$ (5). By Corollary 4.5.4.

THEOREM 4.6.2. For $L \subseteq \Sigma^o \Sigma^o$, the following conditions are equivalent.

- (1) L is in $\mathcal{NB}_2$.
- (2) L is a finite union of sets of the form ${}^aXYZ^a$, where X , Y and Z are regular languages.
- (3) There exists L' in $\mathcal{N}_2(\Sigma^o \times \Sigma^o)$ such that $L = \pi(L')$.
- (4) L is in $\mathcal{NB}_6$.
- (5) L is in $\mathcal{DB}_2$.

PROOF. (1) $\Rightarrow$ (2). Let $L = B_2(M)$. Then $L = \pi(L_2(M))$ and by Propositions 4.5.1 and 4.5.3, $L_2(M)$ is a finite union of sets of the form (UV^a, WZ^a) such that U, V, W and Z are regular languages. If we set $X = V^o$ and $Y = U^o W$, then $\pi(UV^a, WZ^a) = {}^aXYZ^a$.

(2) $\Rightarrow$ (3). Since $\mathcal{N}_2(\Sigma^o \times \Sigma^o)$ is closed under union, it suffices to show that (2) implies (3) for a set of the form $L = {}^aXYZ^a$ such that X, Y and Z are regular languages. Let $\langle M_-; s_- \rangle$ be an *nfa* such that $L_2(\langle M_-;$

$s_- \rangle) = (X^a)^a$ and $\langle M_+; s_+ \rangle$ be an *nfa* such that $L_2(\langle M_+; s_+ \rangle) = YZ^a$. If we choose $H = \{(s_-, s_+)\}$, then the *nba* $M = \langle M_-, M_+, H \rangle$ has the property that $L_2(M) = ((X^a)^a, YZ^a)$ and $L = \pi(L_2(M))$.

(3) $\Rightarrow$ (1). By bilateralization which preserves the condition C_2 .

(1) $\Leftrightarrow$ (4) $\Leftrightarrow$ (5). By Corollary 4.5.4.

THEOREM 4.6.3. *For $L \subseteq \Sigma^* \Sigma^*$, the following conditions are equivalent.*

- (1) L is in $\mathcal{NB}_3$.
- (2) L is of the form $\Sigma^* X \Sigma^*$, where X is a regular language.
- (3) L is in $\mathcal{DB}_3$.

PROOF (1) $\Rightarrow$ (2). Let $L = B_3(M)$. Then $L = \pi(L_3(M))$ and $L_3(M)$ is a finite union of sets of the form $(U \Sigma^*, V \Sigma^*)$ such that U and V are regular languages. If we set $Y = U^a V$ and let X be union of Y 's, then $\pi(U \Sigma^*, V \Sigma^*) = \Sigma^* Y \Sigma^*$ and $\pi(L_3(M)) = \Sigma^* X \Sigma^*$.

(2) $\Rightarrow$ (1). Let $L = \Sigma^* X \Sigma^*$ such that X is a regular language. Let $M = \langle S, \Sigma, \Delta, s_0, F \rangle$ be a *dfa* such that $X = L(M)$. Then evidently we have $X \Sigma^* = L_3(M)$. From M we construct three *nba*'s M_1 , M_2 and M_3 such that:

$$\begin{aligned} L_3(M_1) &= (\Sigma^*, \Sigma^* X \Sigma^*) , \\ L_3(M_2) &= (\Sigma^* X^a \Sigma^*, \Sigma^*) , \\ L_3(M_3) &= \{(ux, vy) \mid u^a v \in X, x \text{ and } y \in \Sigma^*\} . \end{aligned}$$

We can construct an *nba* M' from M_1 , M_2 and M_3 such that $L_3(M') = \bigcup_{i=1}^3 L_3(M_i)$. It is easily seen that M' is bilateral and $L = \pi(L_3(M'))$.

(1) $\Leftrightarrow$ (3). By Corollary 4.5.4.

The set $\Sigma^* \times \Sigma^*$ of biwords is naturally ordered by

$$(u, v) \leq (u', v') \text{ if and only if } u \leq u' \text{ and } v \leq v',$$

where $\leq$ is the relation 'is an initial segment of' over Σ^* . For an increasing sequence (u_n, v_n) of biwords, the supremum denoted by $\sup(u_n, v_n)$ equals to $(\sup(u_n), \sup(v_n))$.

A sequence (u_n, v_n) of biwords is said to be strictly increasing if for all n in N : if u_n is in Σ^* then $u_n = u_{n+1}$ else $u_n < u_{n+1}$; the same holds for v_n . For a language $L \subseteq \Sigma^*$, we associate a two-sided infinite language $L^c = \{\pi(\sup(u_n, v_n)) \mid (u_n, v_n) \text{ are strictly increasing sequences of biwords such that } u_n^a v_n \in L \text{ for all } n\}$.

THEOREM 4.6.4 [10]. *For $L \subseteq {}^\omega \Sigma^\omega$, the following conditions are equivalent.*

- (1) *L is in $\mathcal{DB}_1$.*
- (2) *L is of the form ${}^e X^e$, where X is a regular language.*

PROOF. See [10].

COROLLARY 4.6.5 [10, 25]. *The two-sided infinite language $L = {}^a a\{a, b\}^a \cup {}^a \{a, b\} a^a$ is not in $\mathcal{DB}_1$.*

PROOF. Suppose that there exists a regular language $X \subseteq \{a, b\}^*$ such that $L = {}^e X^e$. For any x in X , ${}^a b x b a^a$ is in L . Therefore there exist $n, m > 0$ such that $a^n b x b a^m$ is in X . If we repeat this process infinitely, we get a two-sided infinite word in L which contains infinitely many b 's in both sides. A contradiction.

We have not yet obtained characterization theorems for the classes $\mathcal{DB}_4 = \mathcal{NB}_4$ and $\mathcal{DB}_6$.

4.7. Concluding remarks

The results obtained in this chapter are summarized in the lattice diagram in Fig. 4-4, where an unbroken line indicates a proper containment and a broken line means a containment whose properness is unknown.

The containment relations of Fig. 4-4 can be proved in the similar way as done in Section 3.5 and thus the details are omitted. For properness we have the following theorem.

THEOREM 4.7.1. *Let $\Sigma = \{a, b\}$. There hold:*

- (1) ${}^\omega \Sigma^\omega \in \mathcal{DB}_3 \cap \mathcal{DB}_4$.
- (2) ${}^a a^\omega \in \mathcal{DB}_4 - \mathcal{DB}_3$.
- (3) ${}^\omega \Sigma b \Sigma^\omega \in \mathcal{DB}_3 - \mathcal{DB}_4$.
- (4) ${}^a a b a^\omega \in \mathcal{DB}_6 - (\mathcal{DB}_3 \cup \mathcal{DB}_4)$.
- (5) ${}^a a \Sigma^\omega \cup {}^\omega \Sigma^1 a^\omega \in \mathcal{DB}_2 - \mathcal{DB}_1$.
- (6) ${}^a (a^* b)^\omega \in \mathcal{DB}_1 - \mathcal{DB}_2$.
- (7) ${}^\omega b (a^* b)^\omega \cup {}^\omega a \Sigma^* b^\omega \in \mathcal{DB}_5 - (\mathcal{DB}_1 \cup \mathcal{DB}_2)$.

PROOF (1). An easy exercise.

(2). We can easily make a dba M such that $\{{}^a a^\omega\} = B_4(M)$. By Theorem

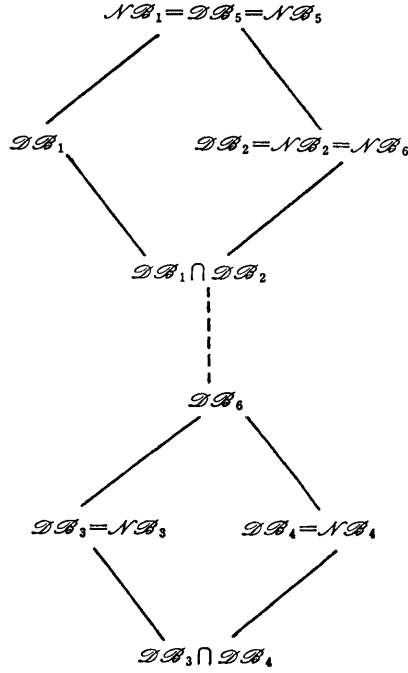


Fig. 4-4

4.6.3, $\{^a a^*\} \notin \mathcal{DB}_3$.

(3). A *dba* M such that $^a \Sigma b \Sigma^a = B_3(M)$ can be easily constructed. Suppose that there exists a *dba* $M = \langle M_-, M_+, H \rangle$ such that $^a \Sigma b \Sigma^a = B_4(M)$. Let $M_- = \langle S_-, \Sigma, \Delta_-, s_{0-}, F_- \rangle$, $M_+ = \langle S_+, \Sigma, \Delta_+, s_{0+}, F_+ \rangle$ and $n = |S_+|$. Since $^a aba^*$ is in $^a \Sigma b \Sigma^a$, $(a^*, a^* ba^*)$ is in $L_4(M)$. Thus there exists (s_-, s_+) in H such that $a^* \in L_4(\langle M_-; s_- \rangle)$ and $a^* ba^* \in L_4(\langle M_+; s_+ \rangle)$. Therefore there exists a C_4 -accepting run r_+ in $Rn(\langle M_+; s_+ \rangle, a^* ba^*)$ such that $O(r_+) \subseteq F_+$. Since $|\{0, \dots, n\}| > n$, for some $i < j \leq n$, $r_+(i) = r_+(j)$. Hence we can make a run r'_+ in $Rn(\langle M_+; s_+ \rangle, a^*)$ such that $r'_+ = r_+(0) \cdot r_+(1) \dots r_+(i-1) \cdot (r_+(i) \dots r_+(j-1))^a$. Clearly $O(r'_+) \subseteq F_+$, and hence $(a^*, a^*) \in L_4(M)$. Since M is bilateral, $^a a^*$ is in $B_4(M)$. A contradiction.

(4). By the proof of Theorem 4.4.1 (Case 5), $^a aba^* \in \mathcal{DB}_6$. By the similar argument in (3) and Theorem 4.6.3, we can show that $^a aba^* \notin \mathcal{DB}_3 \cup \mathcal{DB}_4$.

(5). We can easily make a *dba* M such that $^a \Sigma a^* \cup ^a \Sigma a^* = B_2(M)$. By Corollary 4.6.5, $^a a \Sigma^a \cup ^a \Sigma a^* \notin \mathcal{DB}_1$.

(6). A construction of a dba M such that ${}^o(a*b)^o = B_1(M)$ is easy. Suppose that there exists a dba $M = \langle M_-, M_+, H \rangle$ such that ${}^o(a*b)^o = B_2(M)$. Let $M_- = \langle S_-, \Sigma, \Delta_-, s_{0-}, F_- \rangle$, $M_+ = \langle S_+, \Sigma, \Delta_+, s_{0+}, F_+ \rangle$ and $n = |S_+| + 1$. Since ${}^o(a^n b)^o$ is in ${}^o(a*b)^o$, $((ba^n)^o, (a^n b)^o)$ is in $L_2(M)$. Thus there exists (s_-, s_+) in H such that $(ba^n)^o \in L_2(\langle M_-; s_- \rangle)$ and $(a^n b)^o \in L_2(\langle M_+; s_+ \rangle)$. Therefore there exists a C_2 -accepting run r_+ in $Rn(\langle M_+; s_+ \rangle, (a^n b)^o)$ such that $I(r_+) \subseteq F_+$. Then there exists m_0 in N such that for all $m \geq m_0$, $r_+(m) \in F_+$. Choose any $k > 0$ such that $k(n+1) \geq m_0$. Since $n > |S_+|$, there are repeated states among $r_+(k(n+1)), \dots, r_+((k+1)(n+1)-1)$ which are transitions caused by consecutive n a 's after k -th b . Assume that $r_+(i) = r_+(j)$ for $k(n+1) < i < j < (k+1)(n+1)$. Then we can get a run r'_+ in $Rn(\langle M_+; s_+ \rangle, (a^n b)^k a^o)$ such that $r'_+ = r_+(0) \dots r_+(k(n+1)) \dots r_+(i-1)(r_+(i) \dots r_+(j-1))^o$. Clearly $I(r'_+) \subseteq F_+$, and hence $((ba^n)^o, (a^n b)^k a^o)$ is in $L_2(M)$. Since M is bilateral, ${}^o(a^n b) a^o$ is in $B_2(M)$. A contradiction.

(7). Let $L = {}^o b(a*b)^o \cup {}^o a \Sigma^* b^o$. By the proof of Theorem 4.4.1 (Case 4), L is in $\mathcal{DB}_5$. In the similar argument for (6), we can show that $L \notin \mathcal{DB}_2$. Assume that there exists a dba M such that $L = B_1(M)$. Then by Theorem 4.6.4, there exists a regular language X such that $L = {}^e X^e$. Since ${}^o ab^o \in {}^e X^e$, one can derive the existence of $n_0, m_0 > 0$ such that $a^{n_0} b^{m_0} \in X$. Consider $k_1 > 0$. From ${}^o ab^{n_0} a^{k_1} b^o \in L$ one can find $m_1 > 0$ and $n_1 > n_0$ such that $a^{n_1} b^{m_0} a^{k_1} b^{m_1} \in X$. In this way we can get sequences $n_0 < n_1 < \dots < n_i < \dots, m_0 < m_1 < \dots < m_i < \dots$ and $k_1 < k_2 < \dots < k_i < \dots$ such that $a^{n_i} b^{m_0} a^{k_1} b^{m_1} a^{k_2} b^{m_2} \dots a^{k_i} b^{m_i} \in X$ for each $i > 0$. This means that ${}^o ab^{n_0} a^{k_1} b^{m_1} a^{k_2} b^{m_2} \dots a^{k_i} b^{m_i} \dots$ is in L . A contradiction.

We have not obtained any results concerning characterization of the classes $\mathcal{NBI}$ and $\mathcal{DB}_i$ for $i=1, \dots, 6$. Research for this direction seems to be interesting.

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